GLOBAL SUBTERRANEAN & HARDENED INFRASTRUCTURE: TECHNICAL, ARCHITECTURAL & INSTITUTIONAL AUDIT

Comprehensive Engineering Classification, Geotechnical Specifications, Equipment Inventories, Statutory Ownership & Custodial Officer Registry Across 59 Strategic Subsurface Complexes

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Executive Architectural Synthesis & Taxonomic Distribution

Subterranean infrastructure serves as the physical bedrock of sovereign continuity of government (COG), strategic nuclear command and control (NC3), monetary bullion custody, transuranic and high-level radioactive waste isolation, astroparticle physics detection, and ultra-long-term archival preservation. Across the 59 facilities audited in this technical register, subsurface engineering exploits specific lithological formations—ranging from Precambrian hornblende-biotite granite and Paleozoic bedded halite (rock salt) to tonalite-granodiorite batholiths, dolomitic limestone, and Arctic permafrost sandstone—to achieve physical attenuation against kinetic, nuclear, electromagnetic, and cosmic-ray flux hazards.

Every facility in this audit is evaluated across six immutable evidentiary dimensions: (1) formal designation, historical codenames, and functional taxonomy; (2) exact geographic coordinates and geological host lithology; (3) legal title, operating agency, and prime management contractors; (4) verified executive officers, military commanders, directors, or statutory custodians on record; (5) structural and geotechnical specifications (vertical overburden depth, meters water equivalent shielding, excavated rock volume, blast door tonnage, overpressure rating, and seismic isolation); and (6) internal life-support, power generation, HVAC/CBRN filtration, and specialized mission equipment.

Taxonomic Category	Count	Representative Host Lithologies	Primary Engineering & Shielding Parameters	Facility Registry Numbers
I. Strategic C3I, NC3 & Continuity of Government (Active)	13	Pikes Peak Granite, Metabasalts, Gneiss, Quartzite, Limestone, Arctic Bedrock	610 m granite overburden; 25-ton blast valves; 1,300 helical shock springs; HEMP (MIL-STD-188-125) Faraday cages; closed-loop CBRN life support	#01–#05, #30–#35, #45–#46
II. Decommissioned / Reserve Cold War C2 & Civil Defense Bunkers	9	Bath Oolitic Limestone, London Clay, Graywacke Slate, Devonian Shale, Alluvial Sand	3 m reinforced concrete shells; cast-iron metro tubing (43 m depth); 100-ton concrete/steel blast doors; Luwa/Dräger NBC air filtration banks	#36–#44
III. Sovereign Bullion Depositories, Central Bank & Ecclesiastical Vaults	7	Manhattan Schist, London Clay, Molasse Sandstone, Gotthard Granite, Reinforced Concrete	90-ton revolving steel cylinder doors; 21-inch drill/torch-resistant composite walls; seismic structural isolation; halon/inert gas fire suppression	#15–#20, #54
IV. Commercial Industrial Subsurface Complexes & Archival Biobanks	5	Bethany Falls Limestone, Permian Hutchinson Halite, Permafrost Sandstone, Wasatch Quartz Monzonite	1.3 million m^2 room-and-pillar mining; -18°C permafrost refrigeration; 14-ton Mosler nuclear vault doors; Piql photosensitive silver-halide film	#06–#10
V. Hardened Alpine & Subterranean Data Centers / Transit Nodes	8	Aar Massif Granite, Stockholm Gneiss, Burlington & Bethany Falls Limestone	Tier III/IV 2N+2 diesel rotary UPS; Maybach submarine diesel generators; 1,900 m Alpine overburden; biometric mantraps & EMP attenuation	#11–#14, #51–#53, #55
VI. Deep Geological Nuclear Waste Repositories & Rock Labs	6	Olkiluoto Migmatitic Gneiss, Permian Salado Halite, Miocene Ignimbrite Tuff, Coral Oolite, Zechstein Salt	KBS-3 copper-cast iron canisters + bentonite buffer at 455 m depth; 655 m creeping salt encapsulation; shaft hoisting & robotic emplacement	#21–#23, #56–#58
VII. Deep Underground Astroparticle Physics & Neutrino Laboratories	6	Triassic Marble, Norite/Granite-Gabbro, Dolomitic Limestone, Amphibolite Schist	Up to 2,400 m overburden (6,720 m.w.e.); cosmic muon flux suppression < 10^{-8} $cm^{-2}s^{-1}$; Class 100-2000 cleanrooms; 50,000-ton ultrapure H_2O Cherenkov tanks	#24–#29
VIII. Private Hardened Survival Complexes & Converted ICBM Silos	4	Muschelkalk Limestone, Reinforced Epoxy-Concrete Vertical Silos, Reinforced Munitions Bunkers	52-m deep Atlas-F silo hardened to 100 psi; 2.7-m thick concrete walls; autonomous hydroponics, diesel/wind microgrids & NBC overpressure	#47–#50

Methodological Note: All data points in this audit are derived exclusively from verifiable primary records: United States Department of Defense (DoD) directives and Military Construction (MILCON) budget books, Code of Federal Regulations (44 CFR Part 15; 32 CFR Part 234), U.S. Department of Energy (DOE) Carlsbad Field Office and NRC licensing dockets, German Federal Company for Radioactive Waste Disposal (BGE) technical reports, Posiva Oy nuclear regulatory submissions, national central bank disclosures (FRBNY, Bank of England, SNB, BIS), national physics institute proceedings (INFN, ICRR University of Tokyo, SNOLAB, Tsinghua University, INR RAS, SDSTA), and corporate property/engineering filings.

Section 1: Master 59-Facility Geotechnical & Institutional Cross-Reference Register

Before examining the granular engineering architecture, internal equipment arrays, and administrative chains of custody for each individual complex, Table 1.1 establishes the unified technical and institutional parameters across all 59 audited facilities in the exact sequence of the audit mandate. Figure 1.1 illustrates the comparative vertical rock overburden across the deepest strategic, scientific, and nuclear isolation complexes in the global inventory.

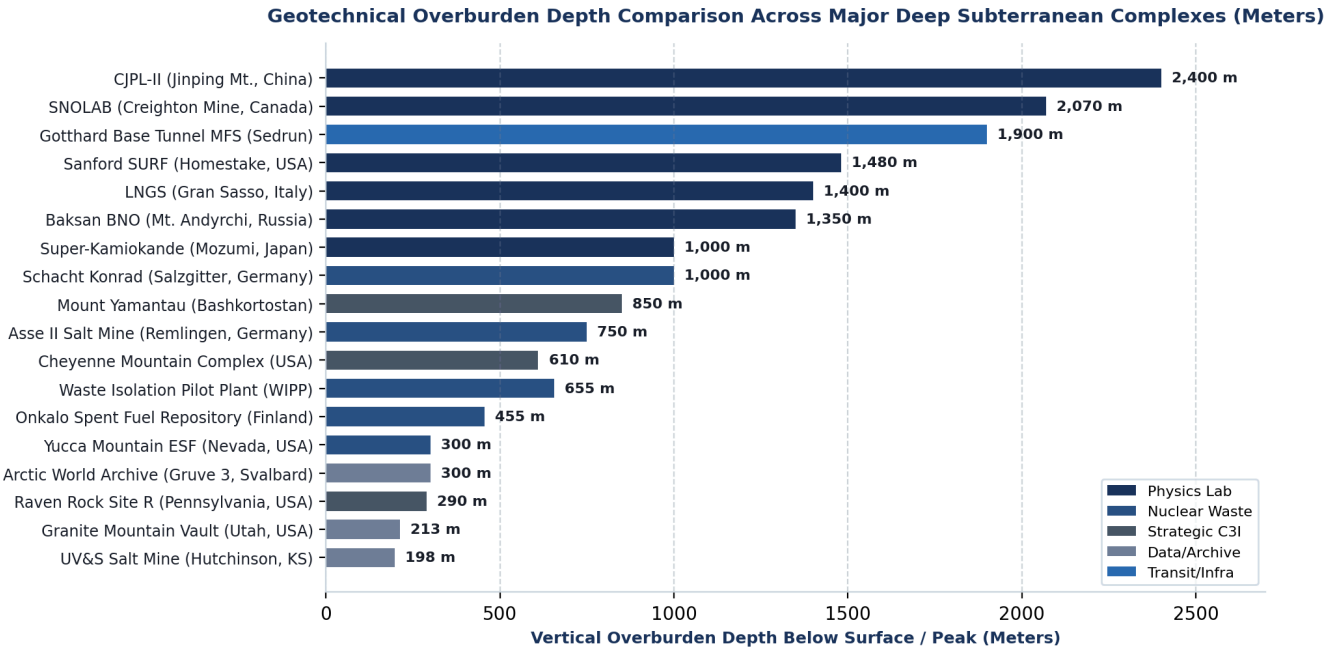


Figure 1.1: Comparative vertical rock overburden depth (in meters below surface or mountain peak) across 18 representative deep geological repositories, astroparticle physics laboratories, strategic C3I bunkers, and archival vaults. Data compiled from Posiva Oy, DOE/WIPP, INFN, Tsinghua University CJPL, SNOLAB, SDSTA, and USNORTHCOM engineering fact sheets.

ID	Facility Name & Designations	Functional Class	Location & Coordinates	Depth, Host Lithology & Scale	Legal Owner & Operating Agency	Registered Commanders, Directors & Custodians
#01	**Cheyenne Mountain Complex (CMSFS / CMOC)**	Strategic C3I / NC3 Alternate	Colorado Springs, CO, USA (38°44'39"N 104°50'48"W)	610 m Pikes Peak Granite; 2.02 ha (5 acres) excavated	U.S. Space Force (Space Base Delta 1) / NORAD & USNORTHCOM	Gen. Gregory M. Guillot (CDR NORAD/USNORTHCOM); Col. Charles J. Cooper (CDR SBD 1)
#02	**Raven Rock Mountain Complex (Site R / AJCC / ANMCC)**	National Military Command / COG	Blue Ridge Summit, PA, USA (39°44'02"N 77°25'10"W)	290 m metabasalt/greenstone; ~65,000 m² across 5 caverns	U.S. Dept. of Defense / Washington Headquarters Services (WHS) / DISA	Michael L. Rhodes (Director, WHS); Lt. Gen. Paul T. Stanton (Director, DISA); Daniel P. Walsh (Director, PFPA)
#03	**Mount Weather Emergency Operations Center (MWEOC / HPSF / Area B)**	Civilian Federal COG / FEMA Hub	Bluemont, VA, USA (39°03'47"N 77°53'20"W)	30–60 m Catoctin metabasalt; 56,000 m² subterranean Area B	U.S. Dept. of Homeland Security (DHS) / FEMA Continuity Directorate	Deanne Criswell / FEMA Administrator; Executive Director, Mt. Weather EOC (44 CFR § 15.2)
#04	**Peters Mountain Facility (AT&T Project Office)**	Hardened Telecom / Intel Relay	Gordonsville, Albemarle Co., VA (38°09'16"N 78°16'00"W)	Hardened subterranean reinforced concrete & quartzite ridge	AT&T Corp. (Federal / National Security Telecommunications Division)	John T. Stankey (CEO, AT&T Inc.); Lance Spencer (Client Executive VP – Defense, AT&T Public Sector)
#05	**Iron Mountain Boyers Facility (WPA-1 / National Underground Vital Records)**	Federal & Commercial Archival Vault	Boyers, Butler Co., PA, USA (41°06'21"N 79°59'58"W)	67 m (220 ft) Vanport Limestone; 70 ha (175 acres) developed	Iron Mountain Inc. (NYSE: IRM); Major Tenant: U.S. OPM Retirement Services	William L. Meaney (President & CEO, Iron Mountain Inc.); Rob Shriver / OPM Associate Director, Retirement Services

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#06	**SubTropolis (Hunt Midwest Underground Business Complex)**	Commercial Subsurface Industrial / Data	Kansas City, MO, USA (39°09'22"N 94°29'16"W)	49 m (160 ft) Bethany Falls Limestone; 1,300,000 m² (14M sq ft) mined	Hunt Midwest Real Estate Development, Inc. (Lamar Hunt Family)	Ora H. Reynolds (President & CEO, Hunt Midwest); Mike Bell (Executive VP, SubTropolis Development)
#07	**Svalbard Global Seed Vault (Svalbard globale frøhvelv)**	Global Agricultural Genetic Backup	Platåberget, Longyearbyen, Svalbard (78°14'09"N 15°29'29"E)	120 m inside permafrost sandstone; 1,000 m² across 3 vault halls	Kingdom of Norway (Ministry of Agriculture & Food) / Statsbygg / NordGen / Crop Trust	Åsmund Asdal (Seed Vault Coordinator, NordGen); Lise Lykke Steffensen (Exec. Dir., NordGen); Stefan Schmitz (Exec. Dir., Crop Trust)
#08	**Arctic World Archive (AWA / Piql / Gruve 3)**	Ultra-Long-Term Optical Data Vault	Mine No. 3 (Gruve 3), Longyearbyen, Svalbard (78°14'26"N 15°26'48"E)	300 m inside Arctic permafrost mountain; acid-free steel vaults	Piql AS & Store Norske Spitsbergen Kulkompani AS / AWA Foundation STI	Rune Bjerkestrand (Managing Director, Piql AS & Chair, Arctic World Archive Foundation STI)
#09	**Granite Mountain Records Vault (LDS FamilySearch Vault)**	Genealogical & Microfilm Archival Vault	Little Cottonwood Canyon, Salt Lake Co., UT (40°34'23"N 111°45'24"W)	213 m (700 ft) Wasatch Quartz Monzonite granite; 6,000+ m² across 6 chambers	The Church of Jesus Christ of Latter-day Saints / FamilySearch International	Steve Rockwood (President & CEO, FamilySearch International); Elder Patrick Kearon / Church Historian & Recorder
#10	**Underground Vaults & Storage (UV&S Hutchinson Salt Mine)**	Deep Halite Archival & Film Repository	Hutchinson, Reno Co., KS, USA (38°01'56"N 97°52'10"W)	198 m (650 ft) Permian Hutchinson Salt Member; 20+ ha developed	Underground Vaults & Storage, Inc. (UV&S)	Jeff Ollenburger (President, UV&S); Lee Spence (Vice Chairman of the Board & Former President, UV&S)
#11	**Deltalis DK2 (Eielen Fort K7 / Kommandoposten K7)**	Alpine Bunker Data Center & Crypto Vault	Attinghausen, Canton Uri, Switzerland (46°51'35"N 8°37'50"E)	300+ m Aar Massif granite; 15,000 m² across 4 subterranean stories	Deltalis SA (formerly leased from armasuisse / Canton Uri)	Frank Harzheim (Managing Director & CEO, Deltalis SA); Board of Directors, Deltalis SA (CHE-115.006.942)
#12	**K20 Kandersteg (Führungsanlage K20 / Federal Council Bunker)**	Swiss Federal Continuity Command Post	Gasterntal, Kandersteg, Canton Bern, Switzerland (46°27'47"N 7°39'56"E)	800+ m Bernese Oberland limestone/granite; 1,000-person capacity	Swiss Confederation (Federal Dept. of Defence VBS / Swiss Armed Forces)	Viola Amherd / Martin Pfister (Head of VBS); Korpskommandant Thomas Süssli (Chief of the Armed Forces)
#13	**Gotthard Sedrun Multifunktionsstelle (MFS Sedrun)**	Deep Alpine Rail Safety & Emergency Node	Sedrun, Canton Graubünden, Switzerland (46°40'50"N 8°46'34"E)	800 m vertical shaft + 1,900 m mountain overburden; 230 m² cross-section caverns	Swiss Federal Railways (SBB CFF FFS) / AlpTransit Gotthard Ltd	Vincent Ducrot (CEO, SBB CFF FFS); Michel Joye (Head of Infrastructure, SBB)
#14	**Pionen White Mountain Data Center (Bahnhof Pionen)**	Hardened Civil Defense Colocation Bunker	Renstiernas gata 35–37, Södermalm, Stockholm, Sweden (59°18'45"N 18°05'06"E)	30 m (100 ft) Stockholm granitic gneiss; 1,200 m² subterranean halls	Bahnhof AB (publ) (Nasdaq First North: BAHN-B)	Jon Karlung (President & CEO, Bahnhof AB); Anders Nilsson (Chairman of the Board, Bahnhof AB)
#15	**Fort Knox Bullion Depository (United States Bullion Depository)**	Sovereign Gold Bullion Vault	Fort Knox, Bullitt/Hardin Co., KY, USA (37°53'00"N 85°57'55"W)	Subterranean 2-level bedrock vault; 467 m³ granite + 3,200 m³ concrete	U.S. Department of the Treasury / United States Mint	Ventris C. Gibson / Kristie McNally (Director / Acting Director, U.S. Mint); Officer in Charge, Fort Knox Depository
#16	**Federal Reserve Bank of New York Underground Vault**	Monetary Gold Custody & Settlement Vault	33 Liberty Street, Manhattan, NY, USA (40°42'30"N 74°00'32"W)	24 m (80 ft) below street / 15 m below sea level in Manhattan Schist	Federal Reserve Bank of New York (FRBNY)	John C. Williams (President & CEO, FRBNY); Sushmita Shukla (First VP & COO, FRBNY)

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#17	**Bank of England Bullion Vaults**	Sovereign & Custodial Gold Vault Network	Threadneedle Street, City of London, UK (51°30'51"N 0°05'18"W)	2–3 levels below street in London Clay; 9 major bullion chambers	Bank of England (HM Treasury owned)	Andrew Bailey (Governor, Bank of England); Victoria Cleland (Executive Director for Banking, Payments & Innovation & Chief Cashier)
#18	**Bank for International Settlements Deep Vault (BIS Basel)**	Central Bank Clearing & Gold Custody Vault	Centralbahnplatz 2, 4051 Basel, Switzerland (47°32'53"N 7°35'30"E)	Multi-level subterranean reinforced concrete vaults beneath BIS Tower	Bank for International Settlements (BIS)	Pablo Hernández de Cos (General Manager, BIS, assumed July 1, 2025); François Villeroy de Galhau (Chair of the Board)
#19	**Swiss National Bank Vault (SNB Bern)**	Sovereign Monetary Gold Depository	Bundesplatz 1, 3003 Bern, Switzerland (46°56'47"N 7°26'42"E)	Deep multi-tiered bedrock & reinforced concrete vaults below Bundesplatz	Swiss National Bank (Schweizerische Nationalbank / SNB)	Martin Schlegel (Chairman of the Governing Board, SNB); Antoine Martin (Vice Chairman, Bern Head Office)
#20	**Archivum Apostolicum Vaticanum Hypogeum (II Bunker)**	Pontifical Subterranean Archival Vault	Cortile della Pigna, Vatican City State (41°54'17"N 12°27'16"E)	2-story subterranean reinforced concrete hypogeum; 31,000 m³ volume, 43 km shelving	Holy See / Archivum Apostolicum Vaticanum	Archbishop Giovanni Cesare Pagazzi (Archivist & Librarian of the Holy Roman Church); Rev. Prof. Rocco Ronzani, O.S.A. (Prefect)
#21	**Onkalo Spent Nuclear Fuel Repository**	Deep Geological High-Level Nuclear Repository	Olkiluoto Island, Eurajoki, Finland (61°14'13"N 21°28'51"E)	430–455 m Paleoproterozoic migmatitic gneiss; 10+ km excavated tunnels	Posiva Oy (60% Teollisuuden Voima Oyj [TVO], 40% Fortum Power & Heat Oy)	Ilkka Poikolainen (President & CEO, Posiva Oy); Janne Mokka (Chairman / Former CEO)
#22	**Waste Isolation Pilot Plant (WIPP)**	Deep Geological Defense Transuranic Repository	Carlsbad, Eddy Co., NM, USA (32°22'18"N 103°47'37"W)	655 m (2,150 ft) Permian Salado Formation halite; 8 disposal panels	U.S. Department of Energy (DOE Carlsbad Field Office); Operator: SIMCO LLC (Bechtel)	Mark Bollinger (Manager, DOE Carlsbad Field Office); Ken Harrawood (President & Program Manager, Salado Isolation Mining Contractors LLC)
#23	**Yucca Mountain Tunnel Complex (Exploratory Studies Facility)**	Decommissioned/Suspended HLW Tunnel Complex	Nye County, Nevada, USA (36°51'10"N 116°25'36"W)	200–300 m Miocene Topopah Spring welded ignimbrite tuff; 7.9 km U-loop + 2.7 km cross-drift	U.S. Department of Energy (DOE Office of Nuclear Energy)	Dr. Kathryn Huff / Mike Goff (Assistant Secretary for Nuclear Energy, U.S. DOE)
#24	**Laboratori Nazionali del Gran Sasso (LNGS)**	Deep Underground Astroparticle Physics Lab	Assergi, L'Aquila, Abruzzo, Italy (42°27'15"N 13°34'28"E)	1,400 m Triassic/Mesozoic dolomitic limestone (3,800 m.w.e.); 180,000 m³ (3 halls)	Istituto Nazionale di Fisica Nucleare (INFN)	Prof. Ezio Previtali (Director, INFN-LNGS); Prof. Antonio Zoccoli (President, INFN)
#25	**Super-Kamiokande (Kamioka Observatory)**	Deep Underground Water Cherenkov Neutrino Lab	Mozumi Mine, Hida, Gifu Prefecture, Japan (36°25'36"N 137°18'37"E)	1,000 m Mount Ikeno gneiss/limestone (2,700 m.w.e.); 41.4 m × 39.3 m cylindrical cavern	Institute for Cosmic Ray Research (ICRR), The University of Tokyo	Prof. Masato Shiozawa (Director, Kamioka Observatory & Spokesperson, Super-K); Prof. Masahiro Teshima / ICRR Director
#26	**SNOLAB (Sudbury Neutrino Observatory Laboratory)**	Ultra-Deep Cleanroom Astroparticle Lab	Vale Creighton Mine, Lively, Greater Sudbury, ON, Canada (46°28'19"N 81°11'12"W)	2,070 m (6,800 ft) Sudbury Igneous Complex norite/gabbro (6,010 m.w.e.); 5,000 m² Class 2000 cleanroom	SNOLAB Institute (Consortium of 5 Canadian Universities); Host: Vale Canada Ltd.	Dr. Jodi Cooley (Executive Director, SNOLAB); Dr. Clarence Virtue / Board Chair
#27	**China Jinping Underground Laboratory (CJPL-I & CJPL-II)**	World's Deepest Underground Physics Lab	Jinping Mountain, Liangshan, Sichuan, China (28°09'10"N 101°42'47"E)	2,400 m Triassic Baishan marble (6,720 m.w.e.); 300,000 m³ across 4 giant halls	Tsinghua University & Yalong River Hydropower Development Co., Ltd. (SDIC)	Prof. Cheng Jianping (Director, CJPL & Chairman, Beijing Normal Univ.); Prof. Yue Qian (Executive Director, CJPL, Tsinghua Univ.); Prof. Kang Kejun

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#28	**Baksan Neutrino Observatory (BNO INR RAS)**	Deep Subterranean Neutrino & Scintillation Lab	Neutrino Village, Elbrus Dist., Kabardino-Balkaria, Russia (43°16'33"N 42°41'25"E)	Up to 1,350 m Mount Andyrchi (4,700 m.w.e.); 4.1 km twin horizontal adits	Institute for Nuclear Research of the Russian Academy of Sciences (INR RAS)	Dr. Valery V. Kuzminov (Head / Director, BNO INR RAS); Dr. Maxim V. Libanov (Director, INR RAS)
#29	**Sanford Underground Research Facility (SURF)**	Deep Underground Neutrino & Dark Matter Lab	Homestake Mine, Lead, Lawrence Co., SD, USA (44°21'12"N 103°45'04"W)	1,480 m (4,850L) Poorman/Homestake Metamorphic Schist (4,300 m.w.e.); >800,000 m³ DUNE caverns	South Dakota Science and Technology Authority (SDSTA) / U.S. DOE Office of Science	Mike Headley (Executive Director, SDSTA & Laboratory Director, SURF); Casey Peterson (Chairperson, SDSTA Board)
#30	**Base aérienne 942 Lyon-Mont Verdun (BA 942)**	Hardened Air Defense C2 & NATO ACCS Node	Mont Verdun, Poleymieux-au-Mont-d'Or, Rhône, France (45°51'12"N 4°47'06"E)	250 m Jurassic limestone + 14 m concrete cap; 16,000 m² across 4 underground levels	French Air and Space Force (Armée de l'Air et de l'Espace) / CDAOA	Général de corps aérien Vincent Cousin / CDR CDAOA; Colonel Laurent (Base Commander, BA 942 Lyon-Mont Verdun)
#31	**Heng Shan Military Command Center (JOCC)**	Strategic National Tri-Service Wartime C2	Jinan Mountain, Dazhi, Zhongshan Dist., Taipei, Taiwan (25°05'20"N 121°32'40"E)	150–200 m sandstone/shale mountain + steel-reinforced concrete liner; multi-level tunnel grid	Ministry of National Defense (MND), Republic of China (Taiwan) / General Staff HQ	Wellington Koo Li-hsiung (Minister of National Defense, ROC); Adm. Mei Chia-shu (Chief of the General Staff, ROC Armed Forces)
#32	**CP TANGO (Command Post Theater Air Naval Ground Operations)**	Combined US-ROK Wartime Theater C3I Bunker	Cheonggyesan, Seongnam, Gyeonggi-do, South Korea (37°25'38"N 127°04'30"E)	200+ m granitic bedrock inside granite mountain; 2-month autonomous NBC life support	United States Forces Korea (USFK) & ROK-US Combined Forces Command (CFC)	Gen. Paul J. LaCamera / Gen. Xavier T. Brunson (Commander, UNC/CFC/USFK); Gen. Kang Shin-chul (Deputy Commander, CFC)
#33	**Gyeryongdae Complex (Bunker B-2 / Joint Chiefs Wartime C2)**	Tri-Service Armed Forces HQ & Subsurface C2	Sindoan-myeon, Gyeryong, South Chungcheong, South Korea (36°18'24"N 127°14'15"E)	Deep granitic mountain tunnels beneath Gyeryongsan; 3-level hardened underground C2	Ministry of National Defense (ROK) / HQ ROK Army, Navy & Air Force	Kim Yong-hyun / Shin Won-sik (Minister of National Defense, ROK); Adm. Kim Myung-soo (Chairman, ROK Joint Chiefs of Staff)
#34	**Mount Yamantau Facility (Mezhgorye / Beloretsk-15 & -16)**	Deep Strategic Continuity & Nuclear C3I Complex	Beloretsky Dist., Bashkortostan, Urals, Russia (54°15'18"N 58°06'07"E)	Up to 850 m Riphean quartzitic sandstone; multi-shaft mine complex operated by ZATO Mezhgorye	Ministry of Defense of the Russian Federation / Directorate of Special Construction (US-30)	Andrei Belousov (Minister of Defense, RF); Viktor Panchenkov / Head of Administration, ZATO Mezhgorye
#35	**Kosvinsky Kamen Complex (Kytlym / Perimetr NC3 Node)**	Strategic Rocket Forces Alternate NC3 & Dead Hand Hub	Karpinsk Dist., Sverdlovsk Oblast, Northern Urals, Russia (59°30'57"N 59°03'43"E)	300 m Paleozoic dunite-pyroxenite-gabbro pluton; ultra-low-frequency VLF/seismic geophone array	Strategic Rocket Forces (RVSN) & 12th Chief Directorate (12 GUMO), Russian MoD	Col. Gen. Sergei Karakayev (Commander, Strategic Rocket Forces RVSN); Lt. Gen. Igor Kolesnikov (Chief, 12th Chief Directorate)
#36	**Objekt 1180 (Olișcani Soviet Reserve Command Bunker)**	Unfinished Theater Strategic C2 Bunker (Abandoned)	Olișcani Forest, Șoldănești Dist., Moldova (47°48'29"N 28°41'38"E)	60 m deep vertical cylindrical concrete shafts (32 m diameter × 12 levels)	State Agency for Military Property / Ministry of Defense of the Republic of Moldova (Unoccupied)	Anatolie Nosatîi (Minister of Defense, Republic of Moldova); Local Municipal Custody, Olișcani
#37	**Bunker 703 (ChZ-703 / Special Archive of the USSR MFA)**	Decommissioned Deep Cryogenic/Archival Bunker (Museum)	2nd Novokuznetsky Lane 14s1, Zamoskvorechye, Moscow, Russia (55°43'55"N 37°38'05"E)	43 m deep in Carboniferous limestone/clay; twin 9.5 m diameter cast-iron metro tubbing blocks	ANO 'Center for Study of Modern Fortification' (formerly Ministry of Foreign Affairs of Russia)	Dr. Dmitry E. Yurkov (Historian, Founder & Scientific Director, Bunker-703 Museum)
#38	**Bunker 42 (Tagansky Protected Command Point / GO-42 / ChZ-293)**	Decommissioned Long-Range Aviation C2 Bunker (Museum)	5th Kotelnichesky Lane 11, Tagansky Dist., Moscow, Russia (55°44'30"N 37°38'54"E)	65 m deep; 7,000 m² across 4 interconnected 9.5 m diameter cast-iron/concrete blocks	OOO 'Kompaniya Novik-Servis' (Private Operator since 2006 federal auction)	Igor A. Kurylev (Owner & General Director, OOO 'Kompaniya Novik-Servis' / Bunker-42 Exhibition Complex)

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#39	**Central Government War Headquarters (Burlington / Site 3 / Turnstile)**	Decommissioned UK National COG Underground City	MoD Corsham, Spring Quarry, Wiltshire, UK (51°25'45"N 2°12'18"W)	30 m Bath oolitic limestone; 97 ha (240 acres), 100 km of roadways, 4,000-person capacity	UK Ministry of Defence (Defence Infrastructure Organisation / Defence Digital at MoD Corsham)	John Healey (Secretary of State for Defence); Mike Green (Chief Executive, Defence Infrastructure Organisation)
#40	**Hack Green Bunker (ROTOR M2 SOC / RGHQ 10.2)**	Decommissioned Radar SOC & Regional Government HQ (Museum)	Baddington Lane, Nantwich, Cheshire, UK (53°02'48"N 2°31'16"W)	Semi-sunken & subterranean 2-level bunker (3,300 m²); 3 m reinforced concrete walls	Hack Green Secret Nuclear Bunker Trust / Siebert Family	Lucy Siebert (Museum Director, Hack Green Secret Nuclear Bunker); Rodney Siebert (Founder & Curator)
#41	**Kelvedon Hatch Secret Nuclear Bunker (ROTOR R4 SOC / RGHQ 5.1)**	Decommissioned 3-Level RGHQ & Radar Bunker (Museum)	Kelvedon Hall Lane, Brentwood, Essex, UK (51°40'12"N 0°15'22"E)	38 m underground in London Clay; 3,500 m² across 3 levels; 1.5-ton blast doors	J.A. Parrish & Sons Ltd. (Private Landowner & Museum Operator)	Mike Parrish (Owner & Managing Director, Kelvedon Hatch Secret Nuclear Bunker)
#42	**Diefenbunker (CEGHQ Carp / CFS Carp)**	Decommissioned Canadian Federal COG Bunker (National Historic Site)	3929 Carp Road, Carp, Ottawa, ON, Canada (45°21'06"N 76°02'50"W)	4-story subterranean reinforced concrete box in alluvial sand/gravel; 9,300 m² (100,000 sq ft)	Diefenbunker: Canada's Cold War Museum (Charitable Not-for-Profit Corp.)	Christine McGuire (Executive Director, Diefenbunker Museum); Susan McLeod (Chair, Board of Directors)
#43	**Bunker D-0 ARK (Atomska Ratna Komanda / Tito's Bunker)**	Decommissioned Yugoslav Supreme Command Bunker (Museum)	Zlatar Mountain, Konjic, Bosnia and Herzegovina (43°38'23"N 17°59'41"E)	280 m overburden beneath Zlatar peak; 6,854 m² U-shaped complex + 12 residential/C2 blocks	Ministry of Defence of Bosnia and Herzegovina; Managed by Agency for Economic Development 'Prvi korak' d.o.o. Konjic	Zukan Helez (Minister of Defence, BiH); Ajla Greda / Director, 'Prvi korak' d.o.o. Konjic
#44	**Dienststelle Marienthal (Regierungsbunker Ahrweiler)**	Decommissioned West German Federal COG Bunker (Museum)	Am Silberberg, Bad Neuenahr-Ahrweiler, RP, Germany (50°32'47"N 7°04'38"E)	110 m Devonian graywacke slate; originally 17.3 km twin railway tunnels (203 m preserved as museum)	Bundesanstalt für Immobilienaufgaben (BImA); Operated by Heimatverein 'Alt-Ahrweiler' e.V.	Executive Board (Vorstand), Bundesanstalt für Immobilienaufgaben (BImA); Karin Joachim (Site Director, Dokumentationsstätte Regierungsbunker)
#45	**Muskö Naval Base (Musköbasen / Muskövarvet)**	Hardened Subterranean Naval Dockyard & Fleet HQ	Muskö Island, Haninge Municipality, Stockholm Archipelago, Sweden (58°59'28"N 18°07'24"E)	40–65 m Precambrian granite; 1.5 million m³ rock blasted; 3 underground drydocks (up to 350 m long) + 20 km roads	Swedish Armed Forces (Försvarsmakten / Marinen) & Saab Kockums AB	Rear Admiral (Konteramiral) Johan Norlén (Chief of the Swedish Navy); Mats Wicksell (Head of Business Area Kockums, Saab AB)
#46	**Haakonsvern Underground Naval Facility**	Hardened Subterranean Submarine Cavern & Fleet Base	Mathopen, Bergen, Vestland, Norway (60°20'16"N 5°13'54"E)	Deep gneiss/granite mountain caverns housing drydock, torpedo/missile magazines, and command center	Royal Norwegian Navy (Sjøforsvaret) & Forsvarsbygg (Norwegian Defence Estates Agency)	Rear Admiral (Kontreadmiral) Oliver Berdal (Chief of the Royal Norwegian Navy); Kommandørkaptein / Base Commander Haakonsvern
#47	**Vivos Europa One (Rothenstein Complex)**	Private Hardened Luxury Survival Complex	Rothenstein, Saale-Holzland-Kreis, Thuringia, Germany (50°50'42"N 11°36'08"E)	Up to 100 m Muschelkalk/Buntsandstein limestone; 21,135 m² (227,904 sq ft) across 5 km of tunnels	The Vivos Group (Terravivos LLC)	Robert K. Vicino (Founder, President & CEO, The Vivos Group)
#48	**Vivos xPoint (Former Black Hills Ordnance Depot)**	Private Hardened Bunker Community (575 Igloos)	Igloo, Edgemont, Fall River Co., SD, USA (43°13'22"N 103°56'28"W)	575 earth-bermed steel-reinforced concrete munitions bunkers across 46.6 km² (18 sq mi)	The Vivos Group (Long-term ground lease from cattle ranch land trust)	Robert K. Vicino (Founder, President & CEO, The Vivos Group)
#49	**Survival Condo (Converted Atlas-F ICBM Silo)**	Hardened Vertical Luxury Condominium Bunker	Near Glasco / Concordia, Cloud Co., KS, USA (39°25'15"N 97°42'10"W)	53 m (174 ft) vertical reinforced epoxy-concrete silo (16.5 m diameter); 1,858 m² across 15 floors	Hardened Structures, LLC / Survival Condo LLC	Larry Hall (Founder, Owner, CEO & Chief Engineer, Survival Condo LLC)

ID	Facility Name & Designations	Functional Class	Location & Coordinates	Depth, Host Lithology & Scale	Legal Owner & Operating Agency	Registered Commanders, Directors & Custodians
#50	**The Oppidum (Kostelec nad Černými lesy Complex)**	Private Ultra-HNWI Subterranean Residence & Vault	Vrchovany / Kostelec nad Černými lesy, Central Bohemia, Czechia (49°59'45"N 14°52'10"E)	Subterranean 7,200 m² (77,500 sq ft) two-level Cold War C2 bunker inside 30,000 m² walled estate	Oppidum Bunkers SA / Jakub Zamrazil Real Estate Holdings	Jakub Zamrazil (Founder, Principal Owner & Managing Director, The Oppidum)
#51	**InfoBunker (Former USAF/AT&T Cold War Bunker)**	Hardened Subterranean Colocation Data Center	3101 Ingersoll Ave (HQ) / Macksburg-Winterset site, IA, USA (41°18'15"N 94°08'40"W)	Subterranean 6,038 m² (65,000 sq ft) reinforced concrete bunker (0.9 m walls/ceiling)	InfoBunker LLC / Five Nines Technology Group	Five Nines & InfoBunker Executive Management Team, Five Nines & InfoBunker Operations
#52	**Bluebird Underground Data Center (BBU)**	Subterranean Limestone Mine Colocation Data Center	2100 E. Trafficway St., Springfield, MO, USA (37°13'12"N 93°15'18"W)	26 m (85 ft) Burlington-Keokuk Limestone; 7,432+ m² (80,000+ sq ft) conditioned white/gray space	Bluebird Network, LLC (Bluebird Fiber, backed by Macquarie Asset Management)	Michael Morey (President & CEO, Bluebird Network / Bluebird Fiber)
#53	**Cavern Technologies Data Center (LightEdge Lenexa)**	Subterranean Limestone Cavern Data Center	17501 W. 98th St., Ste 18-39, Lenexa, KS, USA (38°57'04"N 94°47'28"W)	38 m (125 ft) Bethany Falls Limestone inside 28 ha (3M sq ft) Meritex Executive Park; >18,500 m² data halls	LightEdge Solutions, Inc. (acquired Cavern Technologies in Sept 2021; GI Partners)	Jim Masterson (CEO, LightEdge Solutions); John Clune (Co-Founder, Cavern Technologies)
#54	**Swiss Gold Safe Gotthard Vault (Gotthard Vault 1 & 2)**	High-Security Private Alpine Bullion & Safe Deposit Vault	Altdorf & Gotthard Massif, Canton Uri, Switzerland (46°52'50"N 8°38'22"E)	Deep Alpine granite caverns & Grade X/XI CEN-certified vaults; bonded & segregated bullion storage	Swiss Gold Safe AG (CHE-113.025.816)	Ludwig Karl (Member of the Board of Directors & Spokesperson, Swiss Gold Safe AG); Board of Directors (Verwaltungsrat), Swiss Gold Safe AG
#55	**Safe Host SH2 (STACK Infrastructure GVA02)**	High-Security Hardened Enterprise Data Center	Av. des Baumettes 23, 1196 Gland, Canton Vaud, Switzerland (46°25'28"N 6°16'04"E)	14,000 m² reinforced concrete multi-level vault structure; 16 MW IT load, Tier III/EN 50600 certified	Safe Host SA / STACK Infrastructure EMEA (acquired by IPI Partners in May 2022)	Gérard Sikias (Co-Founder & Former CEO, Safe Host SA); John Eland (CEO, STACK Infrastructure EMEA)
#56	**Olkiluoto Research Tunnel (VLJ Research Tunnel)**	Underground Rock Characterization & KBS-3 Test Facility	Olkiluoto Island, Eurajoki, Finland (61°14'05"N 21°26'25"E)	60–100 m depth adjacent to VLJ LLW/ILW repository in tonalitic gneiss; full-scale deposition holes	Teollisuuden Voima Oyj (TVO) & Posiva Oy	Jarmo Tanhua (President & CEO, Teollisuuden Voima Oyj); Ilkka Poikolainen (President & CEO, Posiva Oy)
#57	**Schacht Konrad (Endlager Konrad)**	Deep Geological Low/Intermediate-Level Waste Repository	Bleckenstedt, Salzgitter, Lower Saxony, Germany (52°11'02"N 10°24'11"E)	800–1,300 m Upper Jurassic (Oxfordian) Coral Oolite iron ore horizon; 303,000 m³ licensed waste volume	Bundesgesellschaft für Endlagerung mbH (BGE) (100% Federal Ministry BMUV)	Iris Graffunder (Chairwoman of the Management Board / CEO, BGE); Marlis Koop & Dr. Thomas Lautsch (Managing Directors, BGE)
#58	**Asse II Mine (Schachtanlage Asse II)**	Deep Salt Mine Nuclear Waste Retrieval & Remediation Site	Remlingen, Wolfenbüttel Dist., Lower Saxony, Germany (52°07'46"N 10°40'16"E)	490–750 m Zechstein halite anticline; 126,000 drums LLW/ILW undergoing mandatory retrieval via Shaft Asse 5	Bundesgesellschaft für Endlagerung mbH (BGE)	Iris Graffunder (Chairwoman of the Management Board, BGE); Dr. Thomas Lautsch (Technical Managing Director, BGE)
#59	**Meishan Command Center (Chiayi / Central-Southern Reserve C2)**	Hardened Regional / Reserve Underground Military C2	Meishan / Dalin Sector, Chiayi Co. & Central Mountain Range, Taiwan (23°35'00"N 120°33'00"E)	Reinforced subterranean rock tunnel command post integrated with ROC Army 10th Corps & reserve C4ISR network	Ministry of National Defense (MND), Republic of China (Taiwan) / ROC Army 10th Army Corps	Wellington Koo Li-hsiung (Minister of National Defense, ROC); Lt. Gen. Chang Yuan-hsun / Commander, 10th Army Corps

Section 2: United States Strategic C3I, Continuity of Government & Vital Records Complexes (#01–#05)

The primary tier of United States national security subterranean infrastructure comprises hardened military command posts, civilian federal continuity hubs, nuclear-survivable telecommunications nodes, and subterranean vital records repositories constructed between 1951 and 1966 and continuously modernized under Department of Defense (DoD), Department of Homeland Security (DHS), and commercial federal contracting frameworks.

#01. Cheyenne Mountain Complex (NORAD Alternate Command Center / CMSFS / CMOC)

Classification & Operational Purpose: Hardened Subterranean Strategic Aerospace Warning, Integrated Tactical Warning and Attack Assessment (ITW/AA), Space Domain Awareness, and National Continuity of Operations (COOP) Command Center (NORAD Fact Sheet; USNORTHCOM). Following the 2006–2008 transition to "warm standby" and subsequent 2015 reactivation of integrated communications architectures under a $700 million Raytheon contract, the complex operates as the fully crewed Alternate Command Center for the North American Aerospace Defense Command (NORAD) and United States Northern Command (USNORTHCOM) (U.S. GAO Report GAO-09-994R).

Location, Lithology & Structural Architecture: Located at 38°44'39"N, 104°50'48"W on the eastern slope of Cheyenne Mountain near Colorado Springs, El Paso County, Colorado. Excavated between May 1961 and 1965 by the Utah Construction & Mining Company under supervision of the U.S. Army Corps of Engineers (Omaha District), removing 693,000 short tons (529,000 m^3) of Precambrian Pikes Peak biotitic granite beneath 610 meters (2,000 feet) of solid rock overburden (Cheyenne Mountain Complex Technical Dossier). Within a 2.02-hectare (5-acre) grid of intersecting chambers (Chambers A, B, C, and D, reaching 18.4 m high and 14 m wide), 15 freestanding three-story low-carbon steel buildings—providing 16,350 m^2 (176,000 sq ft) of floor space—are mounted on 1,319 giant helical steel springs (each weighing 454 kg / 1,000 lbs, fabricated from 7.6 cm diameter steel wire) paired with hydraulic shock absorbers to limit vertical and horizontal shock acceleration during a multi-megaton close-in nuclear detonation (NORAD Fact Sheet).

Equipment, Hardening & Life Support Systems:

- **Blast & HEMP Protection:** Access through the 1.4-km North/South horizontal vehicular tunnel is sealed by two sequential 23-metric-ton (25-short-ton), 1.07-meter-thick steel blast doors engineered to hydraulic closure within 20 seconds (or 45 seconds manually) and rated to withstand a 30-megaton thermonuclear detonation at a 2-kilometer standoff. All 15 internal steel buildings form welded electromagnetic pulse (HEMP) Faraday shields compliant with MIL-STD-188-125.
- **Autonomous Power, Water & CBRN Filtration:** Six 1,750 kW (10.5 MW aggregate) diesel-electric generators backed by a 1,900,000-liter (500,000-gallon) subterranean diesel fuel reservoir and massive uninterruptible battery banks. Four excavated rock reservoirs hold 5.7 million liters (1.5 million gallons) of industrial cooling water and 15.1 million liters (4.0 million gallons) of potable water fed by a natural mountain spring. Overpressure CBRN filtration cascades remove chemical, biological, and radiological particulates, while automated blast valves seal intake and exhaust plenums instantaneously upon shockwave detection by external piezoelectric sensors.

Ownership, Custodianship & Registered Officials: Owned by the United States Department of the Air Force / United States Space Force and administered as **Cheyenne Mountain Space Force Station (CMSFS)** under **Space Base Delta 1 (SBD 1)**, headquartered at Peterson Space Force Base (Space Base Delta 1 Official Roster). Key officials on record:

- **General Gregory M. Guillot, USAF** — Commander, North American Aerospace Defense Command (NORAD) and U.S. Northern Command (USNORTHCOM).
- **Colonel Charles J. Cooper, USSF** — Commander, Space Base Delta 1 (Installation Commander overseeing Peterson SFB and Cheyenne Mountain SFS) (SBD 1 Leadership).
- **Colonel Donald Bartholomew, USSF** — Deputy Commander, Space Base Delta 1; **Chief Master Sgt. Sarah Williams** (Command Senior Enlisted Leader); **Julie A. Cook** (Director of Staff, SBD 1).

#02. Raven Rock Mountain Complex (Site R / AJCC / RRMC / ANMCC / NMCC-R)

Classification & Operational Purpose: Hardened Subterranean Alternate National Military Command Center (ANMCC / NMCC-R), Defense Information Systems Agency (DISA) Computer Operations Megacenter, and Department of Defense Continuity of Operations (COOP) Headquarters (DoD Instruction 5110.11; WHS FY2022 MILCON Budget Justification). Governed under 32 CFR Part 234 ("Conduct on the Pentagon Reservation and the Raven Rock Mountain Complex"), Site R serves as the underground counterpart to the Pentagon's National Military Command Center.

Location, Lithology & Structural Architecture: Situated at 39°44'02"N, 77°25'10"W inside Raven Rock Mountain in Liberty Township, Adams County, Pennsylvania (near Blue Ridge Summit, PA and Fort Ritchie, MD). Excavated beginning in 1951–1953 inside Precambrian Catoctin formation metabasalt (greenstone) and rhyolite beneath 290 meters (950 feet) of rock overburden. The subterranean complex encompasses five major parallel rock-cut caverns (Caverns A through E), housing five freestanding three-story reinforced concrete and steel buildings (Buildings A, B, C, D, and E) plus a dedicated underground power plant chamber, totaling approximately 65,000 m^2 (700,000 sq ft) of hardened subterranean workspace across a 289-hectare (716-acre) reservation.

Equipment, Hardening & Life Support Systems:

- **Blast Doors & Shock Isolation:** Four primary portal tunnels (West, East, North, South) protected by 34-ton steel blast doors and CBRE decontamination airlocks. Internal buildings are structurally isolated from cavern rock walls and ceilings to prevent spalling damage.
- **Power, Cooling & C4ISR:** Six 1,000 kW diesel generators (upgraded under WHS MILCON power-plant modernization programs), subterranean chilled-water reservoirs holding millions of gallons of water, redundant fiber-optic trunks, MILSTAR/AEHF EHF satellite communications terminals on the summit ridge, and the DISARaven Rock Defense Enterprise Computing Center (DECC) alongside Joint Staff, Army (USAISEC), Navy, and Air Force emergency operations centers.

Ownership, Custodianship & Registered Officials: Owned by the U.S. Department of Defense and administered as part of the Pentagon Reservation by **Washington Headquarters Services (WHS)** under DoD Instruction 5110.11, with law enforcement and physical security executed by the **Pentagon Force Protection Agency (PFPA)** (PFPA Annual Report). Key officials on record:

- **Michael L. Rhodes** — Director of Administration and Management (DA&M) and Director, Washington Headquarters Services (WHS), Office of the Secretary of Defense (exercising statutory authority over RRMC per DoDI 5110.11).
- **Daniel P. Walsh** — Director, Pentagon Force Protection Agency (PFPA) (overseeing PFPA Raven Rock Detachment law enforcement and CBRN security).
- **Lt. Gen. Paul T. Stanton, USA** — Director, Defense Information Systems Agency (DISA) and Commander, Department of Defense Cyber Defense Command (overseeing DISA Site R strategic communications nodes).

#03. Mount Weather Emergency Operations Center (MWEOC / High Point Special Facility / Area B)

Classification & Operational Purpose: Primary Civilian Federal Continuity of Government (COG) Complex, National Emergency Coordinating Center, and FEMA National Radio System (FNARS) Master Station (44 CFR Part 15). Operates dual above-ground ("Area A", 176 hectares / 434 acres with 50+ administrative and training buildings) and deep subterranean ("Area B") facilities designed to house senior executive branch leadership, Cabinet succession personnel, and FEMA continuity cadres during National Continuity Policy (NSPD-51 / HSPD-20 / PPD-40) emergencies.

Location, Lithology & Structural Architecture: Located at 39°03'47"N, 77°53'20"W at 19844 Blue Ridge Mountain Road, Bluemont, straddling Loudoun and Clarke Counties, Virginia. Initially established as a U.S. Weather Bureau kite/balloon station in 1902 and converted into an underground experimental test adit by the U.S. Bureau of Mines in 1936, the complex was expanded between 1954 and 1958 by the U.S. Army Corps of Engineers under codename "Operation High Point" in Precambrian Catoctin greenstone/epidosite metabasalt. Area B comprises roughly 56,000 m² (600,000 sq ft) of excavated tunnels and freestanding multi-story office buildings under 30 to 60 meters of rock roof reinforced by more than 40,000 iron rock bolts spaced on 1.2–1.8 meter centers.

Equipment, Hardening & Life Support Systems:

- **Portal Blast Protection & Internal Infrastructure:** The East and West tunnel portals are guarded by a vertical guillotine drop-gate and a 34-ton, 3-meter-high by 6-meter-wide steel blast door with a 2.5-meter-thick concrete radiation plug. Area B contains an underground hospital, mass dormitories, cafeterias, an internal electric transit system, broadcast studios (Emergency Alert System / Primary Entry Point integration), and high-capacity mainframes.
- **Autonomous Utilities:** Subterranean diesel power plant, a 950,000-liter (250,000-gallon) underground potable water reservoir supplemented by surface storage ponds, independent sewage treatment plant, and positive-pressure NBC filtration arrays.

Ownership, Custodianship & Registered Officials: Owned and operated by the **Federal Emergency Management Agency (FEMA)** within the U.S. Department of Homeland Security (DHS), governed by statutory regulations at 44 CFR Part 15 ("Conduct at the Mt. Weather Emergency Assistance Center and at the National Emergency Training Center") (44 CFR § 15.2). Key officials on record:

- **Deanne Criswell / FEMA Administrator** — Administrator, Federal Emergency Management Agency, U.S. Department of Homeland Security.
- **Executive Director, Mount Weather Emergency Operations Center** — Senior Executive Service (SES) official exercising statutory site authority under 44 CFR § 15.2 and § 15.15, assisted by the Mt. Weather Police Force and the Associate Administrator for the Office of National Continuity Programs (ONCP).

#04. Peters Mountain Facility (AT&T Hardened Project Office / Communications Relay)

Classification & Operational Purpose: Hardened Subterranean Telecommunications Switching Center and National Security / Intelligence Continuity Facility (one of five Cold War AT&T Long Lines "Project Offices" constructed around the greater Washington, D.C. / Virginia strategic perimeter alongside Big Hole in Bedford County, VA; Lovettsville/Short Hill Mountain, VA; Warm Springs, VA; and Chatham/White Oak Mountain, VA) (Center for Land Use Interpretation [CLUI] Database; Project Offices Archival Record).

Location, Lithology & Structural Architecture: Located at 38°09'16"N, 78°16'00"W off Peters Mountain Road near Gordonsville and Stony Point in northeastern Albemarle County, Virginia, within the Southwest Mountains ridge system. Constructed in the mid-1960s and substantially expanded in the late 1970s and 1980s inside Catoctin metabasalt and Antietam quartzite formations. The multi-story subterranean reinforced concrete structure is buried beneath the mountain ridge with hardened surface entry portals, security fencing, a helipad, and surface microwave/satellite radomes.

Equipment, Hardening & Life Support Systems: Engineered to survive nuclear overpressure and HEMP disturbances to maintain continuity of government and intelligence data circuits (historically supporting the National Communications System [NCS] and federal intelligence agencies). Equipped with subterranean gas-turbine and diesel generators, chilled-water heat rejection cooling towers concealed within surface baffles, SONET/DWDM optical transport switches, and CBRN positive-pressure air handling systems.

Ownership, Custodianship & Registered Officials: Owned by **AT&T Corp. / AT&T Inc.** (recorded in Albemarle County land and building permit registries under American Telephone & Telegraph Co. / AT&T Communications of Virginia) and operated by AT&T Global Public Sector Solutions for federal continuity contracts (CLUI). Key officials on record:

- **John T. Stankey** — Chairman, President, and Chief Executive Officer, AT&T Inc.
- **Lance Spencer** — Client Executive Vice President – Defense, AT&T Public Sector and FirstNet.

#05. Iron Mountain Boyers Facility (WPA-1 / National Underground Vital Records Center)

Classification & Operational Purpose: Subterranean Limestone Mine Archival Repository, Federal Personnel Records Processing Center, and Enterprise Underground Data Center (Facility Code: WPA-1) (Iron Mountain Official Specification; U.S. OPM Official Report). Houses federal agency records (most notably the U.S. Office of Personnel Management [OPM] Retirement Operations Center storing over 26,000 file cabinets of federal civil service retirement files), alongside Corbis/Bettmann photographic negatives, Universal Music Group master recordings, patented pharmaceutical samples, and the Iron Mountain WPA-1 Tier III colocation data center.

Location, Lithology & Structural Architecture: Located at 1137 Branchton Road, Boyers, Butler County, Pennsylvania (41°06'21"N, 79°59'58"W). Excavated beginning in 1902 by the Pittsburgh Limestone Company (a subsidiary of U.S. Steel) as a room-and-pillar limestone mine extracting the Pennsylvanian-age Vanport Limestone stratum. Acquired by National Underground Storage in 1954 and merged into Iron Mountain Incorporated in 1998. The complex lies 67 to 70 meters (220 to 230 feet) underground, spanning a total subterranean property of over 400 hectares (1,000 acres), of which approximately 70 hectares (175 acres / 1.8 million sq ft) are developed with paved underground roadways, steel-girder roof reinforcement, shotcrete-lined vaults, and freestanding climate-controlled buildings (Federal Records Operations Analysis).

Equipment, Hardening & Life Support Systems:

- **Geotechnical & Environmental Control:** Natural ambient rock temperature of 11°C–13°C (52°F–55°F), augmented by desiccant dehumidification systems maintaining ANSI/NAPM IT9.11 archival standards (temperature down to -20°C and 20%–35% RH for master film/tape vaults).
- **Data Center Infrastructure (WPA-1):** 14.5 MW utility power capacity, N+1/2N diesel backup generators, geothermal subterranean heat sink cooling loop utilizing an underground mine lake, VESDA laser smoke detection, clean-agent gas fire suppression, and multi-factor biometric security checkpoints staffed by armed guards. Around 2,700 personnel (including ~600 OPM employees and contractors) work inside the subterranean complex daily.

Ownership, Custodianship & Registered Officials: Owned and operated by **Iron Mountain Incorporated (NYSE: IRM)**; federal retirement operations managed by the **U.S. Office of Personnel Management (OPM)** (Iron Mountain Leadership). Key officials on record:

- **William L. Meaney** — President and Chief Executive Officer, Iron Mountain Incorporated (Iron Mountain Governance).
- **Mark Kidd** — Executive Vice President and General Manager, Data Centers & Asset Lifecycle Management, Iron Mountain Incorporated.
- **Rob Shriver / OPM Acting Director & Associate Director of Retirement Services** — U.S. Office of Personnel Management (overseeing the Boyers Retirement Operations Center).

Section 3: Commercial Subterranean Industrial Complexes, Global Biobanks & Archival Vaults (#06–#10)

#06. SubTropolis (Hunt Midwest Underground Business Complex)

Classification & Operational Purpose: World's Largest Commercial Subterranean Industrial, Logistics, Archival, and Data Center Complex (Hunt Midwest SubTropolis Official Specification). Houses more than 55 corporate and federal tenants employing over 1,600 personnel, including the U.S. Postal Service Stamp Fulfillment Services Center, the National Archives and Records Administration (NARA) Central Plains Regional Federal Records Center, the U.S. Environmental Protection Agency (EPA) Region 7 logistics warehouse, automotive upfitting plants (Ford F-150 / Transit commercial upfitting inside SubTropolis Automotive Alley), pharmaceutical storage, and the SubTropolis Technology Center (STC) data center (SubTropolis Technical Record).

Location, Lithology & Structural Architecture: Located at 8300 NE Underground Drive, Kansas City, Clay County, Missouri (39°09'22"N, 94°29'16"W) in the Missouri River bluffs. Created via active room-and-pillar mining by Hunt Midwest Enterprises (founded by NFL Kansas City Chiefs owner Lamar Hunt) starting in 1964 within the 270-million-year-old Pennsylvanian Bethany Falls Limestone stratum at a depth of up to 49 meters (160 feet) below the surface. The total excavated area exceeds 1,300,000 m² (14,000,000 sq ft) with over 930,000 m² (10,000,000 sq ft) of leasable industrial space and continues to expand by roughly 13,000 m² (140,000 sq ft) annually through ongoing limestone aggregate mining (Hunt Midwest Sustainability Report).

Equipment & Engineering Specifications: Supported by a regular grid of 7.6-meter (25-foot) square limestone pillars spaced on 19.8-meter (65-foot) centers, leaving 12.2-meter (40-foot) wide corridors and 4.9-meter (16-foot) clear ceiling heights. Features 13.2 km (8.2 miles) of illuminated paved multi-lane subterranean highways, 3.4 km (2.1 miles) of Norfolk Southern and KCS underground rail spurs, 500+ tractor-trailer loading docks, natural 18°C–21°C (65°F–70°F) thermal stability reducing HVAC energy loads by 50%–70%, and high-density fiber-optic trunks.

Ownership, Custodianship & Registered Officials: Owned and operated by **Hunt Midwest Real Estate Development, Inc.** (a privately held Hunt Family enterprise) (Hunt Midwest Leadership). Key officials on record:

- **Ora H. Reynolds** — President and Chief Executive Officer, Hunt Midwest Enterprises, Inc. (Official Bio).
- **Mike Bell** — Executive Vice President, Commercial & Industrial Real Estate Development, Hunt Midwest.
- **Clark Hunt** — Chairman of the Board, Hunt Midwest Enterprises, Inc.

#07. Svalbard Global Seed Vault (Svalbard globale frøhvelv)

Classification & Operational Purpose: Sovereign Permafrost Subterranean Agricultural Germplasm Backup Facility ("Doomsday Vault") (Crop Trust Official Portal; NordGen Technical FAQ). Provides a duplicate safety backup of global food and forage crop diversity under "black-box" deposit agreements (depositors retain sole ownership of their seed accessions). As of June 2026, the facility secures more than 1,400,000 distinct seed samples representing over 6,000 plant species deposited by more than 120 national and international genebanks (SeedVault.no June 2026 Milestone Report).

Location, Lithology & Structural Architecture: Located at 78°14'09"N, 15°29'29"E inside Platåberget (Plateau Mountain) near Longyearbyen on Spitsbergen Island in the Svalbard archipelago, Norway, at an elevation of 130 meters (430 feet) above sea level—ensuring the site remains dry even under complete polar ice-sheet melt scenarios. Excavated in 2006–2008 and upgraded in 2018–2020 (a NOK 200 million Statsbygg waterproof concrete access-tunnel reconstruction replacing the original steel-pipe entrance tunnel following permafrost meltwater intrusion in 2016). A 100-meter

horizontal waterproofed reinforced concrete tunnel leads to three separate rock-cut vault caverns (each measuring 27 m × 10 m × 6 m) situated under 120 meters (390 feet) of Triassic-Jurassic sandstone and shale permafrost overburden.

Equipment, Refrigeration & Storage Capacity:

- **Capacity:** Each of the three vault chambers holds up to 1.5 million seed accessions (4.5 million accessions total, equating to roughly 2.25 billion individual seeds). Seeds are vacuum-sealed in three-ply heat-sealed aluminum foil packets inside standardized 60 × 40 × 28 cm polypropylene boxes on corrosion-resistant steel shelving across Vault Halls 1, 2, and 3.
- **Cryogenic/Permafrost Cooling:** Dual CO_2 transcritical refrigeration compressors (powered by Longyearbyen local grid with dedicated backup generators) chill the active vault chambers to -18°C (-0.4°F). If all mechanical power fails, the surrounding -3°C to -4°C Arctic permafrost guarantees sub-freezing temperatures for decades.

Ownership, Custodianship & Registered Officials: Owned by the **Kingdom of Norway (Ministry of Agriculture and Food)**, constructed and maintained as physical real property by **Statsbygg** (Norwegian Directorate of Public Construction and Property), and jointly operated under a tripartite agreement with the **Nordic Genetic Resource Center (NordGen)** and the **Global Crop Diversity Trust (Crop Trust)** (Crop Trust Governance). Key officials on record:

- **Åsmund Asdal** — Svalbard Global Seed Vault Coordinator, Nordic Genetic Resource Center (NordGen).
- **Lise Lykke Steffensen** — Executive Director, Nordic Genetic Resource Center (NordGen) (NordGen Leadership).
- **Dr. Stefan Schmitz** — Executive Director, Global Crop Diversity Trust (Crop Trust).
- **Harald Nikolaisen** — Director General (Administrerende direktør), Statsbygg.

#08. Arctic World Archive (AWA / Piql / Gruve 3)

Classification & Operational Purpose: Ultra-Long-Term Subterranean Offline Optical Data Repository for Global Cultural, Scientific, and Open-Source Software Heritage (Piql Official Press Release). Opened on March 27, 2017, AWA stores irreplaceable digital archives from over 30 nations—including digitized manuscripts of the Vatican Apostolic Library, the National Archives of Brazil and Mexico, the European Space Agency (ESA), the Edvard Munch Museum, and the **GitHub Arctic Code Vault** (21 TB of open-source software repositories encoded across 186 reels of piqlFilm).

Location, Lithology & Structural Architecture: Located at 78°14'26"N, 15°26'48"E inside the decommissioned **Gruve 3 (Mine No. 3)** coal mine in Platåberget near Longyearbyen, Spitsbergen, Svalbard, Norway (adjacent to, but structurally separate from, the Svalbard Global Seed Vault). The secure steel archive vault is positioned roughly 300 meters (980 feet) horizontal/vertical distance inside the permafrost mountain within a demilitarized zone governed by the Svalbard Treaty of 1920 (AWA Technical Overview).

Equipment & Archival Media Technology: Data is written offline onto **piqlFilm**—a high-resolution photosensitive silver-halide polyester 35mm optical film coated with protective layers and encoded with high-density 2D QR-style binary matrices (storing ~120 GB per reel) alongside human-readable decoding instructions and metadata frames. The reels are sealed inside acid-free, EMP-immune polymer and steel cassettes engineered for a verified 500-to-2,000-year lifespan at ambient permafrost temperatures (-4°C to -5°C) with zero electricity consumption.

Ownership, Custodianship & Registered Officials: Operated jointly by **Piql AS** (Drammen, Norway) and state-owned **Store Norske Spitsbergen Kulkompani AS**, and governed since January 2026 by the **Arctic World Archive Foundation STI** (AWA Foundation Announcement). Key officials on record:

- **Rune Bjerkestrand** — Founder & Managing Director, Piql AS; Chairman of the Board, Arctic World Archive Foundation STI (Piql Leadership).
- **Rune Midtgaard** — Chief Executive Officer (Administrerende direktør), Store Norske Spitsbergen Kulkompani AS.

#09. Granite Mountain Records Vault (LDS FamilySearch Vault)

Classification & Operational Purpose: Hardened Subterranean Genealogical Microfilm and Digital Master Repository (Church of Jesus Christ Newsroom Dossier). Houses the world's largest collection of genealogical and vital historical records: over 2.4 million rolls of 35mm/16mm silver-halide master microfilm, 1 million microfiches, and petabyte-scale high-density digital tape/server arrays containing over 3.5 billion images of parish registers, civil vital statistics, census rolls, and probate records from more than 110 countries (FamilySearch Infrastructure).

Location, Lithology & Structural Architecture: Located at 40°34'23"N, 111°45'24"W at 6695 Little Cottonwood Canyon Road, Salt Lake County, Utah, approximately 24 km (15 miles) southeast of Salt Lake City. Excavated between 1958 and 1963 (completed in 1965 at a cost of $2 million) into the north wall of Little Cottonwood Canyon within the Oligocene Little Cottonwood Stock—a massive, homogeneous batholith of quartz monzonite (granite). Three parallel access tunnels (one central vehicular/loading tunnel and two flanking pedestrian/utility tunnels) penetrate 180 meters into the cliff face underneath 213 meters (700 feet) of solid granite overburden. Three transverse cross-tunnels connect six colossal storage chambers, each measuring 58 meters (190 feet) long, 7.6 meters (25 feet) wide, and 4.6 meters (15 feet) high, providing over 6,000 m² (65,000 sq ft) of hardened vault floor space.

Equipment, Blast Doors & Environmental Systems:

- **Nuclear Blast Doors:** The three main vault entrances are sealed by custom Mosler Safe Company nuclear blast doors—a 14-ton (12.7-metric-ton) central door and two 9-ton (8.2-metric-ton) flanking doors—engineered to withstand a nuclear shockwave overpressure in excess of 500 psi.
- **Climate, Power & Optical Conversion:** Corrugated steel chamber liners with multi-layer waterproof membranes channel natural spring water (~110,000 liters/day) into a dedicated potable storage reservoir. Natural ambient rock temperature remains 13°C–14°C (55°F–57°F) year-round with 35%–40% relative humidity, supplemented by HEPA/carbon filtration, autonomous diesel generators, and high-speed optical microfilm-to-digital scanning arrays linked by redundant fiber optics to Salt Lake City data centers.

Ownership, Custodianship & Registered Officials: Owned by the **Corporation of the President of The Church of Jesus Christ of Latter-day Saints** and operated by **FamilySearch International** and the Church History Department (FamilySearch Governance). Key officials on record:

- **Steve Rockwood** — President and Chief Executive Officer, FamilySearch International.
- **Elder Kyle S. McKay** — Church Historian and Recorder, The Church of Jesus Christ of Latter-day Saints.

#10. Underground Vaults & Storage (UV&S Hutchinson Salt Mine)

Classification & Operational Purpose: Deep Subterranean Bedded Halite Archival, Motion Picture Master Film, and Critical Data Repository (UV&S Official Specification). Founded in 1959, UV&S stores millions of physical archival items, corporate vital records, medical records, magnetic tapes, and original Hollywood camera negatives and studio master film reels (including major motion picture studios such as Warner Bros., MGM, Disney, Sony Pictures, and Paramount—housing original masters of *The Wizard of Oz*, *Gone with the Wind*, and *Star Wars*) alongside the adjacent Strataca (Kansas Underground Salt Museum) cavern sector.

Location, Lithology & Structural Architecture: Located at 3300 E. Avenue G (shaft headframe) / 906 N. Halstead St. (corporate HQ), Hutchinson, Reno County, Kansas (38°01'56"N, 97°52'10"W). Situated 198 meters (650 feet) vertically underground inside the Permian Wellington Formation (Hutchinson Salt Member)—a 275-million-year-old bedded rock salt deposit approximately 100 meters thick, actively mined via room-and-pillar methods by the Hutchinson Salt Company since 1923 (UV&S Corporate Record). UV&S leases and operates more than 20 hectares (50+ acres) of excavated salt bays (each roughly 91 m × 15 m × 3.4 m high, supported by massive solid halite pillars).

Equipment, Access & Environmental Characteristics: All personnel and materials descend 198 meters via a double-deck mine hoist cage (capacity ~3,175 kg / 7,000 lbs) in 75 seconds through a concrete-lined vertical shaft, creating an inherent physical choke-point security barrier. Because rock salt is naturally hygroscopic and geologically impermeable to groundwater, the mine maintains a constant natural environment of 20°C (68°F) and 40%–45% relative humidity without mechanical power, supplemented by enclosed refrigerated cold-vault modules (-18°C / 0°F to 4°C / 40°F at 30% RH) for cellulose nitrate and acetate color film preservation.

Ownership, Custodianship & Registered Officials: Owned and operated by **Underground Vaults & Storage, Inc. (UV&S)** (privately held Kansas corporation; mine horizon owned by Hutchinson Salt Company) (UV&S Leadership Team). Key officials on record:

- **Jeff Ollenburger** — President, Underground Vaults & Storage, Inc. (UV&S) (appointed May 2023) (UV&S Leadership Announcement).
- **Lee Spence** — Vice Chairman of the Board and Former President, Underground Vaults & Storage, Inc.
- **UV&S Executive Operations & Security Team** — Director of Operations and Security, UV&S Hutchinson.

Section 4: Hardened European & Alpine Data Centers, Command Bunkers & Deep Transit Nodes (#11–#14)

#11. Deltalis DK2 (Eielen Fort K7 / Kommandoposten K7)

Classification & Operational Purpose: Hardened Alpine Military Bunker Converted into High-Security Enterprise Colocation and Cryptocurrency Cold-Storage Vault (Facility Code: DK2 / Eielen Fort K7) (Data Center Map Specification; Data Center Dynamics). Constructed during the Cold War as *Kommandoposten K7*, the subterranean headquarters and command center of the Swiss Air Force (Schweizer Luftwaffe) and Gotthard Redoubt defense sector capable of housing 1,500 troops; demilitarized in the 2000s, acquired by Deltalis SA in 2010, and transitioned from retail colocation into dedicated private institutional and sovereign digital asset custody in 2018 (Inside-IT Swiss Registry).

Location, Lithology & Structural Architecture: Located at Eielen Fort DK2, 6468 Attinghausen (near Altdorf), Canton Uri, Switzerland (46°51'35"N, 8°37'50"E). Excavated deep inside the Aar Massif granite beneath more than 300 meters of Alpine rock overburden. The complex spans 15,000 m² (161,458 sq ft) across four subterranean stories connected by a 300-meter-long horizontal vehicular access tunnel protected by sequenced multi-ton Swiss Army nuclear blast doors.

Equipment, Power & Security Systems: Tier IV-equivalent fault-tolerant power fed by dual Swiss hydroelectric utility substations, 2N diesel rotary UPS generators housed in isolated blast chambers, natural Alpine mountain groundwater cooling loops, MIL-STD HEMP shielding, Luwa NBC overpressure air filtration, and five sequential biometric security zones.

Ownership, Custodianship & Registered Officials: Operated by **Deltalis SA** (Swiss Commercial Register CHE-115.006.942, headquartered in Attinghausen/Zug) (Reuters Dossier). Key officials on record:

- **Frank Harzheim** — Managing Director and Chief Executive Officer, Deltalis SA (DCD Report).
- **Board of Directors, Deltalis SA** — Attinghausen, Canton Uri, Switzerland.

#12. K20 Kandersteg (Führungsanlage K20 / Swiss Federal Council Bunker)

Classification & Operational Purpose: Active Strategic Continuity of Government (Regierungsbunker) and Supreme Military Command Center of the Swiss Federal Council (Bundesrat) and Swiss Armed Forces General Staff (Führungsanlage K20 Technical Record; Der Spiegel Analysis). Built to replace the WWII-era *K8 Brindlistollen* (Uri) and Cold War *K10 "Alpenrösli"* (Brienz) federal bunkers, K20 serves as the primary hardened command post and sovereign data node of the Swiss Confederation.

Location, Lithology & Structural Architecture: Located at 46°27'47"N, 7°39'56"E at the entrance to the Gasterntal valley along the Kander River near Kandersteg, Bernese Oberland, Canton Bern, Switzerland, adjacent to the Lötschberg railway tunnel axis. Excavated between 1987 and 1999 at a capital cost of CHF 238 million (with an annual maintenance budget of approximately CHF 350,000–500,000) deep beneath 800+ meters of

Balmhorn/Doldenhorn Alpine limestone and crystalline rock overburden. Designed to house up to 1,000 personnel (the seven members of the Swiss Federal Council, the Federal Chancellor, the Chief of the Armed Forces, and essential C4ISR staff) autonomously for up to six months.

Equipment, Hardening & Life Support Systems: Equipped with multi-stage hydraulic nuclear blast doors, shock-mounted steel-frame command modules, full HEMP shielding, independent subterranean hydroelectric/diesel power generation, autonomous water treatment, NBC regeneration filters, and encrypted links to the Swiss Armed Forces *Führungsnetz Schweiz* (Swiss Command Network) and Onyx SIGINT stations.

Ownership, Custodianship & Registered Officials: Owned by the **Swiss Confederation** and operated by the **Federal Department of Defence, Civil Protection and Sport (VBS / DDPS)** and the **Command Support Organisation (Führungsunterstützungsbasis [FUB] / Kommando Cyber)** of the Swiss Armed Forces. Key officials on record:

- **Federal Councillor Viola Amherd / Martin Pfister** — Head of the Federal Department of Defence, Civil Protection and Sport (VBS).
- **Korpskommandant (Lt. Gen.) Thomas Süssli** — Chief of the Swiss Armed Forces (Chef der Armee).
- **Divisionär (Maj. Gen.) Alain Vuitel / Chief of Cyber Command (Kommando Cyber)** — Custodial operator of Swiss Armed Forces strategic subterranean C4ISR nodes.

#13. Gotthard Sedrun Multifunktionsstelle (MFS Sedrun / Gotthard Base Tunnel)

Classification & Operational Purpose: Deep Subterranean Railway Multifunction Station, Emergency Evacuation Cavern Complex, Ventilation Hub, and High-Voltage Optical/Power Node within the 57.09-km Gotthard Base Tunnel—the world's longest and deepest railway tunnel (Implenia Lot 360 Engineering Specification; Gotthard Base Tunnel Technical Profile).

Location, Lithology & Structural Architecture: Located at 46°40'50"N, 8°46'34"E directly beneath Sedrun (Tujetsch) in Canton Graubünden, Switzerland. Constructed between 1996 and 2016 under Lot 360 by the Transco Sedrun consortium (led by Implenia) for AlpTransit Gotthard Ltd at a depth of **800 meters vertically below the valley floor of Sedrun** (and up to 1,900–2,300 meters below the surrounding Alpine peaks of the Gotthard Massif) within the complex Tavetsch Intermediate Massif (granite, gneiss, and squeezing phyllite/schist). Accessed from the surface via a 1-km horizontal adit leading to two 800-meter-deep vertical blind shafts (Shaft I: 8.6 m diameter equipped with a 50-ton double-winding Koepe hoist travelling at 16 m/s; Shaft II: 7.0 m ventilation/auxiliary shaft) (Sika Gotthard Base Tunnel Engineering Monograph).

Equipment & Subterranean Engineering Systems:

- **Cavern Geometry:** Comprises longitudinal and transverse equipment caverns, four high-speed railway crossover caverns (excavation cross-section up to 230 m²) connecting the East and West single-track tubes, two 450-meter-long pressurized emergency passenger evacuation stations (capable of evacuating 1,000+ passengers into a smoke-free positive-pressure refuge area), and 22 km of interconnected galleries at the base of the shafts.
- **Ventilation, Cooling & Telecoms:** Massive axial exhaust and fresh-air supply fans moving 500 m³/s of air down the 800-m vertical shafts; subterranean water-chiller heat exchangers maintaining tunnel ambient temperatures below 28°C (rock virgin temperature reaches 45°C); redundant 132 kV/15 kV traction substations; and GSM-R/ETCS Level 2 signaling and dark-fiber optical backbones across the Alpine North-South corridor.

Ownership, Custodianship & Registered Officials: Owned by the **Swiss Confederation** and operated by **Swiss Federal Railways (SBB CFF FFS)** (transferred from builder AlpTransit Gotthard Ltd upon commissioning in June/December 2016) (SBB Infrastructure Maintenance). Key officials on record:

- **Vincent Ducrot** — Chief Executive Officer (Vorsitzender der Konzernleitung), Swiss Federal Railways (SBB CFF FFS).
- **Michel Joye / Peter Kummer** — Head of Infrastructure (Leiter Infrastruktur), Swiss Federal Railways (SBB CFF FFS).

#14. Pionen White Mountain Data Center (Bahnhof Pionen)

Classification & Operational Purpose: Hardened Subterranean Civil Defense Nuclear Bunker Converted into a High-Security Commercial Colocation Data Center and Network Operations Center (Data Centre Magazine Technical Profile; Hirsch Security Analysis). Originally constructed in 1943 during World War II as the *Pionen* civil defense command bunker in the Katarinaberget / Vita Bergen ("White Mountains") ridge and expanded during the Cold War to withstand a hydrogen bomb detonation; acquired by Swedish ISP Bahnhof AB and redesigned in 2007–2008 by architect Albert France-Lanord (who modeled the interior on production designs by Ken Adam for James Bond films and *Silent Running*). Notably hosted WikiLeaks servers in 2010.

Location, Lithology & Structural Architecture: Located at Renstiernas gata 35–37 in the Södermalm district of central Stockholm, Sweden (59°18'45"N, 18°05'06"E). Buried 30 meters (100 feet) beneath solid Precambrian Stockholm granitic gneiss bedrock. Over 4,000 m³ of additional granite was blasted out in 2007–2008 to yield 1,200 m² (12,900 sq ft) of multi-level subterranean server halls, suspended glass conference spheres, artificial waterfalls, and hydroponic greenhouse lighting systems.

Equipment, Power & Cooling Specifications:

- **Blast Door & Power Plant:** Sealed behind a 40-centimeter (15.7-inch) thick hardened steel nuclear blast door and biometric mantrap. Backup electrical power is generated by two original **MTU / Maybach 12-cylinder diesel engines** designed for German Type 205/206 submarines, producing 1.5 MW of emergency backup power with compressed-air starting systems and subterranean diesel tanks.
- **Cooling & Connectivity:** Baltimore Aircoil chilled-water heat exchangers capable of handling 1.5 MW of thermal load across hundreds of high-density server racks; triple-redundant carrier-neutral fiber-optic entry paths connected directly to the Stockholm Internet Exchange (Netnod).

Ownership, Custodianship & Registered Officials: Owned and operated by **Bahnhof AB (publ)** (Swedish corporate identity number 556519-9493, listed on Nasdaq First North Growth Market: BAHN-B). Key officials on record:

- **Jon Karlung** — President and Chief Executive Officer (VD), Bahnhof AB (Jon Karlung Technical Interview).
- **Anders Nilsson** — Chairman of the Board of Directors (Styrelseordförande), Bahnhof AB.

Section 5: Sovereign Bullion Depositories, Central Bank Deep Vaults & Ecclesiastical Bunkers (#15–#20)

#15. Fort Knox Bullion Depository (United States Bullion Depository)

Classification & Operational Purpose: Primary Sovereign Gold Bullion Depository of the United States Department of the Treasury (U.S. Mint Official Specification; U.S. Mint Historical & Technical Dossier). Constructed in 1936 for $560,000 under the Public Works Administration to relocate monetary gold from coastal New York and Philadelphia vaults inland. Holds 147.34 million fine troy ounces (4,583 metric tons) of U.S. Treasury gold bullion bars (roughly 56.3% of total U.S. official gold reserves, carried at a statutory book value of $42.2222/oz or ~$6.22 billion, and a market value exceeding $380 billion), and historically safeguarded the original U.S. Declaration of Independence, U.S. Constitution, Bill of Rights, Abraham Lincoln's Gettysburg Address, the Gutenberg Bible, and the Magna Carta during World War II.

Location, Lithology & Structural Architecture: Located at 37°53'00"N, 85°57'55"W at Gold Vault Road and Bullion Boulevard adjacent to the U.S. Army Fort Knox installation in Bullitt and Hardin Counties, Kentucky. The fortress building measures 32 m × 37 m (105 ft × 121 ft) and rises 12.8 m (42 ft) above grade, anchored directly onto Mississippian limestone bedrock. Construction consumed 465 m³ (16,500 cu ft) of North Carolina granite lining the outer walls, 3,200 m³ (4,200 cu yd) of concrete, 680 metric tons (750 short tons) of reinforcing steel, and 608 metric tons (670 short tons) of structural steel.

Equipment, Vault Engineering & Security Systems:

- **The Subterranean Two-Level Steel Vault:** At the center of the building sits a two-story subterranean and ground-level vault measuring 12.2 m × 18.3 m (40 ft × 60 ft), structurally independent from the outer granite/concrete shell so that guards can patrol the entire perimeter corridor between the outer walls and the vault box, including viewing underneath and above the vault via mirror periscopes. The vault walls are 53 to 69 cm (21 to 27 inches) thick, composed of laminated steel plates, steel I-beams, and steel cylinders laced with torch- and drill-resistant alloys encased in concrete.
- **Mosler Vault Door:** The main vault door—manufactured by the Mosler Safe Company—is 53 cm (21 inches) thick and weighs over **18 metric tons (20 short tons)**, with a vault casing bringing total entryway assembly weight to ~22 tons. It is sealed by a 100-hour four-movement chronometer time lock and multiple independent combination locks requiring separate Depository staff members to dial their respective combinations (no single individual knows the full sequence).
- **Defense Perimeter:** Surrounded by four corner machine-gun pillboxes, an entrance guard box, triple-fenced high-security intrusion detection zones, seismic ground sensors, thermal/infrared optics, autonomous power and water plants, and the adjacent U.S. Army Fort Knox garrison.

Ownership, Custodianship & Registered Officials: Owned by the **United States Department of the Treasury** and operated by the **United States Mint**, protected by the **United States Mint Police (Protection Division)** (U.S. Mint Police). Key officials on record:

- **Scott Bessent / Janet L. Yellen** — Secretary of the Treasury, U.S. Department of the Treasury.
- **Ventris C. Gibson / Kristie McNally** — Director / Acting Director, United States Mint.
- **Officer in Charge (OIC), United States Bullion Depository at Fort Knox** — Career U.S. Mint Senior Official and U.S. Mint Police Field Commander, Fort Knox Protection Division (U.S. Mint).

#16. Federal Reserve Bank of New York Underground Vault (33 Liberty Street Gold Vault)

Classification & Operational Purpose: World's Largest Monetary Gold Custody and Settlement Vault (FRBNY Official Gold Vault Specification; FRBNY Speech by President John C. Williams). Designed by Architects York & Sawyer with vault engineering by Frederick S. Holmes and completed in 1924, the subterranean vault stores approximately **507,000 standard London Good Delivery gold bars (~6,331 metric tons)** held in earmarked custody for roughly 36 foreign central banks, monetary authorities, international organizations (such as the IMF and BIS), and a small fraction (~5%) of U.S. Treasury reserves.

Location, Lithology & Structural Architecture: Located at 33 Liberty Street (bounded by Liberty, Nassau, William, and Maiden Lane), Financial District, Manhattan, New York, NY (40°42'30"N, 74°00'32"W). Situated on Basement Level E, **24.4 meters (80 feet) below street level and 15.2 meters (50 feet) below sea level**, resting directly on the metamorphic **Manhattan Schist bedrock**—the only geological stratum in Lower Manhattan capable of supporting the concentrated load of over 6,300+ metric tons of gold plus the vault structure without subsidence (FRBNY Building Specification).

Equipment, Revolving Cylinder Door & Custody Compartments:

- **90-Ton Revolving Cylinder Entryway:** Rather than a hinged vault door, the main vault entrance consists of a vertical **2.7-meter (9-foot) tall, 81.6-metric-ton (90-short-ton) solid steel cylinder** rotating inside a **127-metric-ton (140-short-ton) steel-and-concrete frame**. When rotated 90 degrees to close, the cylinder drops 9.5 mm (3/8 inch) onto a tapered seat, forming an airtight, watertight seal capable of withstanding hydrostatic pressure and explosive charges, locked via dual combinations and quadruple time locks.
- **122 Earmarked Compartments:** Inside the three-story subterranean vault complex, gold bars are segregated inside **122 individual steel-bar cages (compartments)**, each assigned to a single account holder (identified solely by numeric code) and secured by a padlock, two separate combination locks (requiring two separate FRBNY groups plus the internal audit staff to open), and an auditor's lead/tamper-evident seal. High-capacity hydraulic platform scales weigh bars to 1/100th of a troy ounce during inter-compartment sovereign settlements.

Ownership, Custodianship & Registered Officials: Owned and operated by the **Federal Reserve Bank of New York (FRBNY)** (Second Federal Reserve District). Key officials on record:

- **John C. Williams** — President and Chief Executive Officer, Federal Reserve Bank of New York (FRBNY).
- **Sushmita Shukla** — First Vice President and Chief Operating Officer, Federal Reserve Bank of New York.
- **Michelle Neal / Head of Markets Group & Central Bank and International Account Services (CBIAS)** — Executive overseeing FRBNY sovereign gold custody operations.

#17. Bank of England Bullion Vaults (Threadneedle Street Underground Vaults)

Classification & Operational Purpose: Sovereign & Commercial Bullion Custody Vault Network of the United Kingdom and the London Bullion Market Association (LBMA) (LBMA Official Technical Profile). The Bank of England vaults constitute the second-largest monetary gold depository on Earth (after FRBNY), storing approximately **400,000 standard 400-troy-ounce (12.4 kg) gold bars (~5,134 to 5,300 metric tons)** on behalf of HM Treasury (the UK Exchange Equalisation Account's ~310.3 metric tons), over 60 foreign central banks, and LBMA commercial clearing banks.

Location, Lithology & Structural Architecture: Located beneath the Bank of England headquarters at Threadneedle Street, City of London, EC2R 8AH, United Kingdom (51°30'51"N, 0°05'18"W). Designed during Sir Herbert Baker's 1925–1939 rebuilding of the Bank (above Sir John Soane's curtain wall), the subterranean substructure extends three stories below ground into the impermeable **London Clay** stratum (formerly the site of the subterranean Walbrook stream). The bullion complex comprises **nine massive subterranean gold vaults** whose total floor area exceeds that of the 42-story Tower 42 (NatWest Tower) skyscraper (Bank of England Vault Specification).

Equipment, Geotechnical Load Limits & Security Systems: Because London Clay is plastic and compressible compared to Manhattan Schist, open-rack pallets inside the nine vaults are stacked to a maximum height of four to six tiers per pallet bay to prevent structural floor settlement. Entry into each vault requires voice-recognition biometrics, electronic keycards, and two physical **91-centimeter (3-foot) long steel keys** operated simultaneously by divorced-custody vault custodians through heavy Chubb steel blast/vault doors. Internal transport uses custom compact electric pallet trucks navigating narrow high-security corridors.

Ownership, Custodianship & Registered Officials: Owned and operated by the **Bank of England** (wholly owned by HM Treasury on behalf of the UK Government). Key officials on record:

- **Andrew Bailey** — Governor of the Bank of England.
- **Victoria Cleland** — Executive Director for Banking, Payments and Innovation, and **Chief Cashier of the Bank of England** (statutory official responsible for Banknotes and Gold Custody operations).
- **Dave Ramsden** — Deputy Governor for Markets and Banking, Bank of England.

#18. Bank for International Settlements Deep Vault (BIS Basel)

Classification & Operational Purpose: International Central Bank Clearing House and Subterranean Sovereign Gold Custody Vault (BIS Governance & Organisation). Established under the Hague Agreements of January 20, 1930, the BIS serves as the "bank for central banks" for 63 member central banks worldwide. While the majority of BIS gold holdings (typically 100 to 500+ metric tons of sight and time gold deposits and swaps) are held on an earmarked or loco basis at FRBNY, the Bank of England, and the Swiss National Bank, the BIS headquarters in Basel maintains multi-level subterranean security vaults beneath the Botta Building and the 1977 Martin Burckhardt BIS Tower. Under Article 10 of the 1930 Hague Constituent Charter and the 1987 Swiss Headquarters Agreement, the BIS buildings, subterranean vaults, and all assets entrusted to them enjoy absolute sovereign immunity from expropriation, requisition, seizure, or attachment in time of peace or war.

Location, Lithology & Structural Architecture: Located at Centralbahnplatz 2, 4051 Basel, Canton Basel-Stadt, Switzerland (47°32'53"N, 7°35'30"E), adjacent to the Basel SBB railway hub. The 18-story, 69.5-meter cylindrical BIS Tower rests on a four-story subterranean reinforced concrete foundation and vault structure excavated into the Rhine Graben alluvial gravel and Tertiary marl/molasse bedrock.

Equipment & Security Systems: Equipped with CEN Grade XI/XII Swiss steel vault doors (manufactured by Kaba / Rüegg / Bauer), seismic vibration arrays, biometric access control, redundant diesel generators, and sovereign cryptographic settlement servers linking the BIS Banking Department directly to the SWIFT, SIC (Swiss Interbank Clearing), Fedwire, and TARGET2 networks.

Ownership, Custodianship & Registered Officials: Owned and operated by the **Bank for International Settlements (BIS)** (BIS Management Roster). Key officials on record:

- **Pablo Hernández de Cos** — General Manager (Chief Executive), Bank for International Settlements (assumed office July 1, 2025, succeeding Agustín Carstens) (BIS Official Profile).
- **François Villeroy de Galhau** — Chairman of the Board of Directors, Bank for International Settlements (and Governor, Banque de France).
- **Andréa M. Maechler** — Deputy General Manager, Bank for International Settlements.
- **Luis Bengoechea** — Head of the Banking Department, Bank for International Settlements.

#19. Swiss National Bank Vault (SNB Bundesplatz 1, Bern)

Classification & Operational Purpose: Primary Sovereign Gold Depository of the Swiss Confederation (SNB Official History). Stores the bulk of Switzerland's official monetary gold reserves—totaling **1,040.0 metric tons** (approximately 33.44 million fine troy ounces) of gold bullion and gold coins, of which **70% (~728 metric tons) is stored domestically in Switzerland** (primarily beneath Bundesplatz 1 in Bern and affiliated Swiss Army Alpine bunkers such as K20/Gotthard), 20% at the Bank of England, and 10% at the Bank of Canada.

Location, Lithology & Structural Architecture: Located at Bundesplatz 1, 3003 Bern, Canton Bern, Switzerland (46°56'47"N, 7°26'42"E), directly across the Federal Square from the Swiss Parliament Building (Bundeshaus). Designed by Bernese architects Eduard Joos and Emil Bracher and

inaugurated on January 20, 1912, the Neo-Renaissance head office sits atop a deep multi-level subterranean vault network excavated into the Bernese Lower Freshwater Molasse (*Berner Sandstein*) bedrock beneath the Bundesplatz (SNB Bern Head Office).

Equipment & Vault Security Specifications: Constructed with multi-meter reinforced concrete walls lined with armor-plated steel, hydraulic watertight security bulkheads (protecting against potential flooding from the nearby Aare River or city water mains), triple-credential access protocols (combined key, combination, and biometric iris/palm verification), ultrasonic/seismic perimeter sensors, and direct tactical response integration with the Bern Cantonal Police and Federal Security Service (Bundessicherheitsdienst).

Ownership, Custodianship & Registered Officials: Owned and operated by the **Swiss National Bank (Schweizerische Nationalbank / SNB)**, a special-statute joint-stock company (SIX: SNBN) governed by the Federal Act on the Swiss National Bank (SNB Governing Board). Key officials on record:

- **Dr. Martin Schlegel** — Chairman of the Governing Board (Präsident des Direktoriums), Swiss National Bank (FSB / SNB Profile).
- **Dr. Antoine Martin** — Vice Chairman of the Governing Board and Head of Department II (Cash, Gold Custody & Financial Stability, headquartered at Bundesplatz 1, Bern).
- **Dr. Petra Tschudin** — Member of the Governing Board and Head of Department III, Swiss National Bank.
- **Barbara Janom Steiner** — President of the Bank Council (Bankratspräsidentin), Swiss National Bank.

#20. Archivum Apostolicum Vaticanum Hypogeum ("Il Bunker" / Vatican Apostolic Archives Underground)

Classification & Operational Purpose: Hardened Subterranean Archival Vault of the Holy See (Archivio Apostolico Vaticano Official Historical & Technical Note). Known until Pope Francis's *Motu Proprio Lux in arcana / L'esperienza storica* of October 22, 2019 as the *Archivum Secretum Vaticanum* (Vatican Secret Archives), the institution safeguards over **85 linear kilometers (53 miles) of shelving** containing papal registers, state correspondence, diplomatic nunciature dispatches, and ecumenical council records spanning twelve centuries (8th to 21st centuries), including the *Dictatus Papae* of Gregory VII (1075), the Bull *Inter caetera* (1493), the 1521 excommunication bull of Martin Luther (*Decet Romanum Pontificem*), King Henry VIII's 1530 Peers' Petition for marriage annulment, and the 1633 Inquisition trial dossier of Galileo Galilei.

Location, Lithology & Structural Architecture: Located at 41°54'17"N, 12°27'16"E directly beneath the **Cortile della Pigna (Courtyard of the Pinecone)** of the Apostolic Palace / Vatican Museums inside Vatican City State. Excavated between 1978 and 1980 under Pope Paul VI and Pope John Paul II (inaugurated by Pope John Paul II on October 18, 1980), the two-story subterranean reinforced concrete hypogeum ("Il Bunker") encompasses a structural volume of **31,000 m³** and adds **43 linear kilometers (27 miles) of compact rolling steel shelving** across two underground levels.

Equipment, Fire Suppression & Climate Control:

- **Structural & Environmental Isolation:** Encased in waterproofed, seismically reinforced concrete membranes isolated from the Vatican Hill tufa/clay water table. Automated HVAC systems maintain archival parchment and paper compartments at a strict 18°C–20°C and 50%–55% relative humidity.
- **The Reserve Chamber (La Riserva):** The innermost high-security chamber within the subterranean vault is a hermetically sealed, fireproof climate room housing the single most precious papal gold-seal privileges (*Aurea Bullae*), illuminated codices, and royal parchment scrolls, protected by inert-gas (oxygen-depletion/nitrogen) fire suppression, laser smoke detectors, and 24-hour Swiss Guard and Corps of Gendarmerie of Vatican City State surveillance.

Ownership, Custodianship & Registered Officials: Sovereign property of the **Holy See (Vatican City State)**, administered by the **Archivum Apostolicum Vaticanum** (Vatican Apostolic Archive Portal). Key officials on record:

- **Archbishop Giovanni Cesare Pagazzi** — Archivist and Librarian of the Holy Roman Church (appointed by the Holy See, succeeding Cardinal José Tolentino de Mendonça / Archbishop Angelo Vincenzo Zani).
- **Rev. Prof. Rocco Ronzani, O.S.A.** — Prefect of the Vatican Apostolic Archive (appointed July 5, 2024, succeeding Bishop Sergio Pagano, B., who served from 1997 to 2024) (Official Appointment Record).

Section 6: Deep Geological Nuclear Waste Repositories & Underground Physics Laboratories (#21–#29)

#21. Onkalo Spent Nuclear Fuel Repository

Classification & Operational Purpose: World's First Licensed Deep Geological Final Disposal Facility for High-Level Spent Nuclear Fuel (Posiva Oy ONKALO Technical Dossier; IAEA Technical Profile). Constructed to permanently isolate up to 6,500 metric tons of uranium (tU) of spent nuclear fuel generated by Finland's Olkiluoto (OL1, OL2, OL3) and Loviisa (LO1, LO2) nuclear power plants for a minimum regulatory isolation period of 100,000 years using the Swedish-Finnish **KBS-3V multi-barrier disposal concept**. Trial operations (Trial Run of Final Disposal, TRFD) commenced in August 2024, with commercial final disposal hole drilling initiated in August 2026 (NucNet August 2026 Technical Report).

Location, Lithology & Structural Architecture: Located at 61°14'13"N, 21°28'51"E on Olkiluoto Island in the municipality of Eurajoki, Satakunta, western Finland. Excavated since 2004 into 1.8-to-1.9-billion-year-old Paleoproterozoic crystalline bedrock (migmatitic gneiss and tonalitic-granodioritic gneiss) to a repository disposal depth of **430 to 455 meters below sea level**. The subsurface network comprises over 10 kilometers of tunnels: a 5.5-km spiral vehicular access ramp (1:10 gradient, 5.5 m wide × 6.3 m high), three vertical concrete-lined shafts (a canister transfer shaft, a personnel shaft, and ventilation shafts extending to 455 m depth), technical service halls at 437 m depth, and horizontal deposition tunnels up to 350 m long (Posiva Operating Principle).

Equipment & KBS-3V Engineered Barrier Specifications:

- **Above-Ground & Shaft Encapsulation Plant:** Fuel assemblies are dried and sealed inside dual-shell **KBS-3 canisters** (up to 5.25 m long, 1.05 m diameter, weighing ~24.5 metric tons loaded) consisting of a nodular cast-iron load-bearing insert surrounded by a 50-mm-thick corrosion-resistant oxygen-free copper outer shell welded shut via friction-stir welding (FSW) and inspected by ultrasonic/radiographic arrays.
- **Subterranean Robotic Emplacement:** Canisters descend the vertical canister shaft via a KONE hoisting system to the 430-m level, where autonomous rail/wheel transfer vehicles deposit each canister into an individual vertical borehole (8 m deep × 1.75 m diameter) drilled into the floor of the deposition tunnel. Each canister is encased in swelling Wyoming sodium-bentonite clay rings, and completed tunnels are backfilled with bentonite blocks and sealed with low-pH concrete plugs.

Ownership, Custodianship & Registered Officials: Owned and operated by **Posiva Oy** (Finnish Business ID 1029258-8; jointly owned 60% by Teollisuuden Voima Oyj [TVO] and 40% by Fortum Power and Heat Oy), regulated by the **Radiation and Nuclear Safety Authority of Finland (STUK)** (Posiva Oy). Key officials on record:

- **Ilkka Poikolainen** — President and Chief Executive Officer (Toimitusjohtaja), Posiva Oy (appointed September 2023) (Posiva Press Release).
- **Petteri Tiippana** — Director General (Pääjohtaja), Radiation and Nuclear Safety Authority (STUK).

#22. Waste Isolation Pilot Plant (WIPP)

Classification & Operational Purpose: United States Deep Geological Repository for Defense-Related Transuranic (TRU) Radioactive Waste (U.S. DOE WIPP Official Portal). Authorized by Public Law 96-164 (1979) and operational since March 26, 1999, WIPP is the only licensed deep geological nuclear repository in the United States, disposing of contact-handled (CH-TRU) and remote-handled (RH-TRU) waste contaminated with alpha-emitting transuranic elements (principally plutonium-239, americium-241, and neptunium-237 with atomic numbers > 92 and half-lives > 20 years) generated by U.S. nuclear weapons research and production complexes (Los Alamos, Savannah River, Idaho National Lab, Hanford, Oak Ridge, Lawrence Livermore). Total statutory capacity under the WIPP Land Withdrawal Act is 175,564 m^3 (6.2 million cu ft) of TRU waste.

Location, Lithology & Structural Architecture: Located at 32°22'18"N, 103°47'37"W in the Chihuahuan Desert, 42 km (26 miles) east of Carlsbad, Eddy County, New Mexico, within a 41.4 km^2 (16-square-mile) federal land withdrawal zone. Excavated **655 meters (2,150 feet) vertically underground** inside the middle of the 610-meter-thick **Permian Salado Formation**—a 250-million-year-old laterally continuous bedded halite (rock salt) horizon free of flowing groundwater. Under lithostatic rock pressure at 655 m depth, the viscoplastic rock salt creeps inward at 5–7.5 cm per year, naturally encapsulating and crushing the waste drums into a solid, impermeable salt monolith within 100–200 years.

Equipment, Shafts, Panels & Ventilation Systems:

- **Five Vertical Shafts & Disposal Panels:** Connected to the surface by five vertical shafts: the Waste Shaft (equipped with a 45-ton hoist container), Salt Handling Shaft, Air Intake Shaft, Exhaust Shaft, and the newly constructed **Shaft #5 (Utility Shaft)** sunk to 655 m depth as part of the Safety Significant Confinement Ventilation System (SSCVS) upgrade. The underground horizon comprises 8 original disposal panels (each consisting of 7 room-and-pillar chambers measuring 91 m long × 10 m wide × 4 m high), with replacement Panels 11 and 12 under excavation.
- **Cask & Emplacement Equipment:** Waste arrives in NRC-certified Type B shipping casks (TRUPACT-II, TRUPACT-III, HalfPACT, and RH-72B). Remote-handled TRU canisters are emplaced horizontally into shielded boreholes drilled into the salt ribs using a laser-guided hydraulic emplacement machine and sealed with concrete shield plugs, while CH-TRU 55-gallon drums and standard waste boxes are stacked in rooms with magnesium oxide (MgO) supersacks placed atop every stack as an engineered chemical buffer to absorb CO_2 and limit actinide solubility. The new SSCVS provides 15,300 m^3/min (540,000 CFM) of HEPA-filtered underground airflow (DOE WIPP Operational Update).

Ownership, Custodianship & Registered Officials: Owned by the **United States Department of Energy (DOE Carlsbad Field Office [CBFO])** and managed and operated under contract by **Salado Isolation Mining Contractors, LLC (SIMCO)**, a Bechtel National-led consortium (with Los Alamos Technical Associates as nuclear safety subcontractor) whose contract was extended in December 2025 through 2030 (Bechtel Press Release; DOE CBFO Leadership). Key officials on record:

- **Mark Bollinger** — Manager, Carlsbad Field Office (CBFO), U.S. Department of Energy (CBFO Official Profile).
- **Ken Harrawood** — President and Program Manager, Salado Isolation Mining Contractors, LLC (SIMCO) (WIPP Permit Docket NM4890139088).

#23. Yucca Mountain Tunnel Complex (Exploratory Studies Facility / ESF)

Classification & Operational Purpose: Deep Subterranean Exploratory Studies Facility (ESF) and Suspended Federal High-Level Radioactive Waste (HLW) and Spent Nuclear Fuel Repository Complex (U.S. NRC Site Characterization Review; National Academies Engineering Dossier). Designated by the U.S. Congress under the Nuclear Waste Policy Amendments Act of 1987 as the sole candidate site for a 70,000-metric-ton heavy metal (MTHM) deep geological repository for commercial spent nuclear fuel and defense vitrified HLW. While the NRC staff completed its 5-volume Safety Evaluation Report (NUREG-1949) in 2014–2015 finding the technical design compliant with post-closure performance standards, federal licensing appropriations were suspended in 2010–2011, leaving the fully excavated tunnel complex in a secure, unstaffed preservation status.

Location, Lithology & Structural Architecture: Located at 36°51'10"N, 116°25'36"W on the boundary of the Nevada Test and Training Range and the Nevada National Security Site (NNSS) in Nye County, Nevada (~140 km northwest of Las Vegas). Unlike saturated crystalline or salt repositories, Yucca Mountain is situated in the **unsaturated (vadose) zone**—200 to 300 meters below the mountain crest and roughly 300 meters *above* the regional water table—within the 12.8-million-year-old Miocene **Topopah Spring Tuff** (a densely welded rhyolitic ignimbrite volcanic ash-flow tuff) (Yucca Mountain Technical Record).

Equipment & Subterranean Tunnel Geometry:

- **Main 7.9-km U-Loop & 2.7-km Cross-Drift:** Between September 1994 and April 1997, a 7.62-meter (25-foot) diameter Robbins Tunnel Boring Machine (TBM) excavated the 7.87-kilometer (4.9-mile) horseshoe-shaped **Exploratory Studies Facility (ESF) Main Loop** from the North Portal to the South Portal, lined with steel sets, rock bolts, welded wire mesh, and precast concrete invert segments. In 1997–1998, a second 5.0-meter

(16.4-foot) diameter TBM excavated the 2.68-kilometer (1.7-mile) **Enhanced Characterization of the Repository Block (ECRB) Cross-Drift** plus nine specialized thermal-hydrological-mechanical-chemical (THMC) test alcoves (Alcoves 1–8 and the Heated Drift test room).

Ownership, Custodianship & Registered Officials: Owned by the **United States Federal Government** (straddling DOE Nevada National Security Site, U.S. Air Force NTTR, and Bureau of Land Management public lands) and under the custodial jurisdiction of the **U.S. Department of Energy (DOE Office of Nuclear Energy)**, with perimeter security maintained by Mission Support and Test Services, LLC (MSTS) at the Nevada National Security Site (NRC EIS Review). Key officials on record:

- **Dr. Mike Goff / Principal Deputy Assistant Secretary for Nuclear Energy** — Office of Nuclear Energy, U.S. Department of Energy.
- **Manager, Nevada Field Office** — National Nuclear Security Administration (NNSA), U.S. Department of Energy.

#24. Laboratori Nazionali del Gran Sasso (LNGS / INFN)

Classification & Operational Purpose: World's Largest By-Volume Horizontal-Access Deep Underground Astroparticle and Nuclear Physics Research Center (INFN-LNGS Official Portal). Conceived in 1979 by INFN President Antonino Zichichi and inaugurated in 1987, LNGS provides an ultra-low-background cosmic-ray environment for dark matter direct detection, neutrinoless double-beta decay, solar and geo-neutrino spectroscopy, and nuclear astrophysics.

Location, Lithology & Structural Architecture: Located at 42°27'15"N, 13°34'28"E beside the 10.2-km Gran Sasso d'Italia A24 freeway tunnel between L'Aquila (Assergi) and Teramo in Abruzzo, Italy. Buried beneath **1,400 meters (4,593 feet) of Triassic/Mesozoic dolomitic and calcareous limestone** (Mount Aquila / Corno Grande massif), providing a cosmic-ray shielding depth of **3,800 meters water equivalent (m.w.e.)**—reducing the cosmic muon flux by a factor of 1,000,000 to approximately 1.1 muons m^{-2} h^{-1} (3.0×10^{-4} m^{-2} s^{-1}), while the low-uranium/thorium carbonate host rock keeps ambient radiogenic neutron flux 1,000 times lower than surface levels. The subterranean laboratory spans **17,800 m² (191,600 sq ft) of floor space and 180,000 m³ of excavated volume**, centered on three monumental experimental halls—**Hall A, Hall B, and Hall C**—each measuring approximately 100 meters long, 20 meters wide, and 18 meters high, linked by horizontal service tunnels directly accessible by heavy highway trucks from the A24 tunnel.

Equipment & Major Experimental Detectors:

- **Dark Matter & Neutrino Arrays:** Houses **XENONnT** (an 8.5-metric-ton dual-phase liquid xenon time projection chamber searching for WIMP dark matter), **DarkSide-20k** (a 50-ton depleted underground liquid argon TPC), **CUORE / CUPID** (Cryogenic Underground Observatory for Rare Events operating 988 TeO_2 bolometric crystals cooled to 10 millikelvin—the coldest cubic meter in the known universe), **GERDA / LEGEND-200** (enriched germanium-76 neutrinoless double-beta decay detector), **Borexino** (300-ton liquid scintillator solar neutrino detector), **LVD** (Large Volume Detector with 1,000 tons of liquid scintillator for core-collapse supernova neutrinos), and the **LUNA-MV** (Laboratory for Underground Nuclear Astrophysics 3.5 MV accelerator).

Ownership, Custodianship & Registered Officials: Owned and operated by the **Istituto Nazionale di Fisica Nucleare (INFN)** (Italian National Institute for Nuclear Physics, supervised by the Ministry of University and Research [MUR]) (INFN-LNGS Director Page). Key officials on record:

- **Prof. Ezio Previtali** — Director, Laboratori Nazionali del Gran Sasso (INFN-LNGS) (serving since October 15, 2020) (LNGS Official Biography).
- **Prof. Antonio Zoccoli** — President, Istituto Nazionale di Fisica Nucleare (INFN).

#25. Super-Kamiokande (Kamioka Observatory / Super-K / Mozumi Mine)

Classification & Operational Purpose: Deep Subterranean 50,000-Metric-Ton Ultrapure Water Cherenkov Neutrino Detector and Proton Decay Observatory (Kamioka Observatory Official Portal). Operational since April 1996, Super-Kamiokande discovered atmospheric neutrino oscillations in 1998 (earning Kajita Takaaki the 2015 Nobel Prize in Physics, following Masatoshi Koshiba's 2002 Nobel Prize for Kamiokande-II's detection of Supernova 1987A neutrinos), serves as the far detector for the T2K (Tokai-to-Kamioka) 295-km long-baseline neutrino beam from J-PARC, and was upgraded in 2020 with dissolved gadolinium sulfate (**SK-Gd**, 0.01% to 0.03% Gd loading) to detect the Diffuse Supernova Neutrino Background (DSNB) (ICRR 30th Anniversary Report).

Location, Lithology & Structural Architecture: Located at 36°25'36"N, 137°18'37"E at a horizontal tunnel distance of 1.7 km inside the active/decommissioned **Mozumi Mine** of the Kamioka Mining and Smelting Co. beneath the peak of Mount Ikeno (Ikenoyama) in Kamioka-cho, Hida City, Gifu Prefecture, Japan. Buried beneath **1,000 meters (3,280 feet) of Hida metamorphic gneiss and crystalline limestone** (equivalent to **2,700 m.w.e.**), the detector occupies a domed cylindrical rock cavern measuring **41.4 meters high and 39.3 meters in diameter**.

Equipment & Photomultiplier Detector Array:

- **50,000-Ton Stainless Steel Tank & 13,000 PMTs:** Welded stainless-steel cylindrical tank (41.4 m high × 39.3 m diameter) holding 50,000 metric tons of continuously recirculated, degassed, UV-sterilized ultrapure water (fiducial inner volume: 22,500 tons). An optical barrier divides the tank into an Inner Detector (ID) lined with **11,129 inward-facing 50-cm (20-inch) Hamamatsu R3600 hemispherical photomultiplier tubes (PMTs)** that detect faint blue Cherenkov light cones emitted by relativistic charged leptons (electrons and muons) produced by neutrino interactions, and an Outer Detector (OD) lined with **1,885 outward-facing 20-cm (8-inch) PMTs** acting as an active cosmic-ray muon veto shield (Masato Shiozawa Technical Profile). Adjacent caverns in the Kamioka complex also house the **KAGRA** 3-km cryogenic underground gravitational-wave laser interferometer, **KamLAND**, and **XMASS**, while the 260,000-ton **Hyper-Kamiokande** cavern is under construction 8 km south under Mount Nijuugo.

Ownership, Custodianship & Registered Officials: Operated by the **Kamioka Observatory, Institute for Cosmic Ray Research (ICRR), The University of Tokyo** (in collaboration with the Super-Kamiokande International Collaboration of ~190 physicists from 40 institutions; mine access infrastructure hosted by Kamioka Mining & Smelting Co., Ltd., a subsidiary of Mitsui Mining & Smelting Co.). Key officials on record:

- **Prof. Masato Shiozawa** — Director, Kamioka Observatory (ICRR, The University of Tokyo) and Spokesperson, Super-Kamiokande Collaboration (ICRR Official Notice).

- **Prof. Teruo Fujii** — President, The University of Tokyo.

#26. SNOLAB (Sudbury Neutrino Observatory Laboratory)

Classification & Operational Purpose: Deepest Operational Class-2000 Cleanroom Underground Astroparticle Physics Laboratory in North America (SNOLAB Official Portal). Expanded from the original Sudbury Neutrino Observatory (SNO—whose heavy-water measurements of solar neutrino flavor oscillations earned Arthur B. McDonald the 2015 Nobel Prize in Physics), SNOLAB specializes in direct dark matter searches, neutrinoless double-beta decay, supernova neutrino detection, and subterranean radiobiology/quantum computing under ultra-low cosmic radiation.

Location, Lithology & Structural Architecture: Located at 46°28'19"N, 81°11'12"W on the 6800-foot level (**2,070 meters / 6,800 feet vertically underground**) of the active **Vale Creighton Nickel-Copper Mine** in Lively, Greater Sudbury, Ontario, Canada. Excavated inside the norite and granite-gabbro footwall of the 1.85-billion-year-old Sudbury Impact Basin. The high density of the norite rock overburden provides a cosmic-ray attenuation of **6,010 meters water equivalent (m.w.e.)**, reducing the residual cosmic muon flux to **0.27 muons m^{-2} day^{-1} (3.1 × 10^{-10} cm^{-2} s^{-1})**—more than 50,000,000 times lower than at the Earth's surface (Jodi Cooley Technical Presentation).

Equipment, Cleanroom Architecture & Detectors:

- **5,000 m^2 Class-2000 Subterranean Cleanroom:** Personnel descend 2 km in the Creighton #9 Mine cage, walk 1.5 km through an active mine drift, and pass through mandatory full-body showers, airlocks, and cleanroom garmenting to enter the **5,000 m^2 (53,800 sq ft) cleanroom complex** (total excavated volume 47,000 m^3, including the 30-m high × 22-m diameter Barrel Cavern, the Cube Hall, and the Cryopit).
- **Active Experiments:** Hosts **SNO+** (an 18-m diameter acrylic vessel filled with 780 metric tons of linear alkylbenzene [LAB] liquid scintillator doped with tellurium-130 inside a 7,000-ton ultrapure water shield), **DEAP-3600** (3.6 tons of liquid argon dark matter detector), **SuperCDMS SNOLAB** (cryogenic germanium/silicon crystal dark matter detectors cooled to 15 mK), **PICO-40L / PICO-500** (superheated C_3F_8 bubble chambers), **NEWS-G**, **DAMIC-M**, and **HALO** (Helium and Lead Observatory for supernova neutrinos).

Ownership, Custodianship & Registered Officials: Operated by the **SNOLAB Institute** (a university consortium governed by Queen's University, Carleton University, Laurentian University, University of Alberta, and Université de Montréal, funded via the Canada Foundation for Innovation [CFI] and Ontario Ministry of Colleges and Universities) within subsurface property hosted by **Vale Canada Limited** (SNOLAB 2025–2026 Annual Report). Key officials on record:

- **Dr. Jodi Cooley** — Executive Director, SNOLAB (and Professor of Physics, Queen's University) (SNOLAB Official Profile).
- **Dr. Clarence Virtue / Board Chair** — Chair of the SNOLAB Institute Board of Directors.

#27. China Jinping Underground Laboratory (CJPL-I & CJPL-II / DURF)

Classification & Operational Purpose: World's Deepest and Largest Operational Deep Underground Physics Laboratory (Tsinghua University CJPL Official Portal; CJPL Technical Dossier). Phase I (CJPL-I, 4,000 m^3) opened on December 12, 2010; Phase II (**CJPL-II / Deep Underground and Ultra-low Radiation Background Facility for Frontier Physics Experiments [DURF]**, 300,000 m^3) entered formal scientific operation in December 2023 as a Chinese National Major Science and Technology Infrastructure facility.

Location, Lithology & Structural Architecture: Located at 28°09'10"N, 101°42'47"E at the midpoint of the 17.5-kilometer Jinping Mountain highway/water-diversion traffic tunnels constructed for the Jinping-I and Jinping-II Hydropower Stations across the Yalong River bend in Liangshan Yi Autonomous Prefecture, Sichuan Province, China. Positioned beneath **2,400 meters (7,874 feet) of vertically overlying Triassic Baishan Formation pure marble**, CJPL achieves a cosmic-ray attenuation depth of **6,720 meters water equivalent (m.w.e.)**—resulting in the lowest residual cosmic-ray muon flux of any laboratory on Earth: **(2.0 ± 0.4) × 10^{-10} muons cm^{-2} s^{-1} (~60 muons m^{-2} $year^{-1}$)**, or less than one hundred-millionth of surface cosmic radiation. Because the surrounding Baishan marble is carbonate rock nearly devoid of uranium-238, thorium-232, and potassium-40 (unlike granitic rocks), natural terrestrial gamma and radon backgrounds are exceptionally low.

Equipment & Monumental Cavern Array:

- **Four 130-Meter Experimental Halls:** CJPL-II comprises four parallel, interconnected cathedral-scale experimental halls (Halls A, B, C, and D), each measuring **130 meters long, 14 meters wide, and 14 meters high**, plus drainage, ventilation, and logistics tunnels totaling **330,000 m^3 of excavated rock volume**—accessible directly via drive-in two-lane highway tunnels.
- **Experimental Payloads:** Houses **PandaX-4T / PandaX-xT** (Particle and Astrophysical Xenon 4-ton to multi-10-ton liquid xenon dark matter and neutrinoless double-beta decay detector led by Shanghai Jiao Tong University), **CDEX-10 / CDEX-100 / CDEX-1T** (China Dark Matter Experiment high-purity germanium detector arrays submerged inside a 1,700 m^3 liquid nitrogen/ultrapure water cryostat tank led by Tsinghua University), **JUNA** (Jinping Underground Nuclear Astrophysics 400 kV high-current ion accelerator), and the **Jinping Neutrino Experiment (JNE)**.

Ownership, Custodianship & Registered Officials: Jointly owned and operated by **Tsinghua University** and **Yalong River Hydropower Development Co., Ltd.** (a subsidiary of SDIC Power Holdings Co., Ltd.) (CJPL Organization Structure). Key officials on record:

- **Prof. Cheng Jianping** — Director, China Jinping Underground Laboratory (CJPL) and Chairman of the University Council, Beijing Normal University / Professor of Engineering Physics, Tsinghua University (Tsinghua Engineering Physics).
- **Prof. Yue Qian** — Executive Director of CJPL, Spokesperson of the CDEX Collaboration, and Professor of Engineering Physics, Tsinghua University (CJPL Faculty).
- **Prof. Kang Kejun** — Founding Director and Chief Scientist of CJPL, Tsinghua University.

#28. Baksan Neutrino Observatory (BNO INR RAS)

Classification & Operational Purpose: First Purpose-Built Deep Subterranean Neutrino and Astroparticle Observatory in the Soviet Union / Russian Federation (INR RAS Official BNO Specification; CERN Courier Technical Profile). Approved by the USSR Council of Ministers in 1966 and operational since 1977 inside the Baksan River Gorge near Mount Elbrus, BNO operates deep radiochemical solar neutrino experiments, scintillation telescopes, gravitational-wave laser interferometers, and low-background geophysics laboratories.

Location, Lithology & Structural Architecture: Located at 43°16'33"N, 42°41'25"E at the dedicated scientific township of **Neutrino (Neytrino)** in the Elbrus District of the Kabardino-Balkarian Republic, North Caucasus, Russia. Consists of two parallel horizontal railway/vehicular adits (Main Adit and Auxiliary Adit) bored **4,100 meters (2.55 miles) horizontally under Mount Andyrchi (peak elevation 3,922 m)** through Paleozoic crystalline schists, gneisses, and plagiogranites, reaching a maximum vertical rock overburden of **1,350 meters (4,700–4,800 meters water equivalent)** (INR RAS BNO Charter).

Equipment & Subterranean Experimental Halls:

- **Baksan Underground Scintillation Telescope (BUST) at 550 m (850 m.w.e.):** Located 550 m from the portal in a 24 × 24 × 16 m cavern; a four-story detector (16.7 × 16.7 × 11.1 m) comprising 3,184 standard aluminum tanks filled with white-spirit liquid scintillator (330 metric tons total) viewed by FEU-49B photomultipliers for atmospheric muon and core-collapse supernova neutrino detection.
- **SAGE & BEST Hall at 3,670 m (4,700 m.w.e.):** A 60 m long × 10 m wide × 12 m high chamber lined with 60 cm of ultra-low-radioactivity dunite concrete (from the Ural Mountains) and 6 mm steel plate, housing the **Soviet-American Gallium Experiment (SAGE)** and the **Baksan Experiment on Sterile Transitions (BEST)** utilizing **50 metric tons of liquid metallic gallium (Ga-71)** irradiated by solar neutrinos and mega-curie Cr-51 / Zn-65 electron-capture neutrino sources to extract individual germanium-71 atoms. Also houses the **OGRAN** 75-m underground opto-acoustic gravitational-wave antenna.

Ownership, Custodianship & Registered Officials: Operated as a branch of the **Institute for Nuclear Research of the Russian Academy of Sciences (INR RAS)** under the Ministry of Science and Higher Education of the Russian Federation. Key officials on record:

- **Dr. Valery V. Kuzminov** — Head / Director of the Baksan Neutrino Observatory Branch (BNO INR RAS) (CERN Courier).
- **Dr. Maxim V. Libanov** — Director, Institute for Nuclear Research of the Russian Academy of Sciences (INR RAS) (Corresponding Member of the RAS).
- **Academician Valery A. Rubakov (deceased) / Academician Vladimir N. Gavrin** — Principal Investigator, SAGE & BEST Gallium Neutrino Experiments, INR RAS.

#29. Sanford Underground Research Facility (SURF / Homestake Mine)

Classification & Operational Purpose: Deepest Underground Scientific Research Laboratory in the United States (SURF Official Portal; U.S. DOE Office of Science Cooperative Agreement). Established inside the former Homestake Gold Mine—where Raymond Davis Jr. operated his 380,000-liter perchloroethylene solar neutrino detector from 1965 to 1994 on the 4850-foot level, earning the 2002 Nobel Prize in Physics—SURF hosts flagship U.S. Department of Energy experiments in dark matter detection, long-baseline neutrino oscillations, nuclear astrophysics, and deep geothermal energy research.

Location, Lithology & Structural Architecture: Located at 630 E. Summit Street, Lead, Lawrence County, South Dakota (44°21'12"N, 103°45'04"W) in the northern Black Hills. The former Homestake Mine (operated 1876–2002 by Homestake Mining Co. / Barrick Gold Corp.) extends across 600 km (370 miles) of drifts down to the 8000-foot level (2,438 m), though active scientific campuses are concentrated on the **4850 Level (1,480 meters / 4,850 feet vertically underground, equivalent to 4,300 m.w.e.)** within Paleoproterozoic Poorman, Homestake, and Ellison Formation iron-magnesium-silicate schists and phyllites. Accessed via two rehabilitated vertical shafts reaching 1,524 m depth: the **Ross Shaft** (steel-lattice headframe, main rock/equipment hoist) and the **Yates Shaft** (personnel/cage hoist).

Equipment, Davis Campus & LBNF/DUNE Cavern Complex:

- **Davis Campus (4850L):** Houses the **LUX-ZEPLIN (LZ)** experiment—a 10-metric-ton (7-ton active) dual-phase liquid xenon time projection chamber inside a 265,000-liter ultrapure water shield holding the world's most sensitive direct dark matter exclusion limits—alongside the **Majorana Demonstrator** (enriched Ge-76 neutrinoless double-beta decay array) and **CASPAR** (Compact Accelerator System for Performing Astrophysical Research 1 MV Van de Graaff accelerator).
- **Long-Baseline Neutrino Facility (LBNF) / DUNE Far Site Caverns (4850L):** Completed excavation in 2024 removing over **800,000 metric tons (~350,000 m³) of hard metamorphic rock** on the 4850 Level to create three colossal caverns—the North Cavern (144.5 m long × 19.8 m wide × 28.0 m high), South Cavern (144.5 m × 19.8 m × 28.0 m), and Central Utility Cavern (190 m × 19.8 m × 11.0 m)—designed to house four **17,500-metric-ton (70,000-ton total) liquid argon time projection chambers (LArTPCs)** for the **Deep Underground Neutrino Experiment (DUNE)** receiving a high-intensity neutrino beam transmitted 1,300 km through the Earth's crust from Fermilab in Batavia, Illinois.

Ownership, Custodianship & Registered Officials: Real property donated by Barrick Gold Corp. in 2006 (augmented by a $70 million gift from philanthropist T. Denny Sanford) to the **South Dakota Science and Technology Authority (SDSTA)** (an agency of the State of South Dakota), operated under a Cooperative Agreement with the **U.S. Department of Energy (DOE Office of Science / Fermi National Accelerator Laboratory)** (SURF Leadership). Key officials on record:

- **Mike Headley** — Executive Director, South Dakota Science and Technology Authority (SDSTA) and Laboratory Director, Sanford Underground Research Facility (SURF) (Official Bio).
- **Casey Peterson** — Chairperson, Board of Directors, South Dakota Science and Technology Authority (SDSTA).
- **Dr. Lia Merminga / Dr. Young-Kee Kim** — Director, Fermi National Accelerator Laboratory (Fermilab, lead DOE laboratory for LBNF/DUNE).

Section 7: Allied, European, Asian & Russian/Soviet Military Command Complexes (#30–#46 & #59)

#30. Base aérienne 942 Lyon-Mont Verdun (BA 942 / Capitaine Robert)

Classification & Operational Purpose: Primary Hardened Subterranean Air Defense Command and Control (C2) Center of the French Air and Space Force and NATO Air Command and Control System (ACCS) Master Radar Node (GlobalMilitary Technical Record; BA 942 Dossier). Houses the Operational Headquarters of the **Commandement de la défense aérienne et des opérations aériennes (CDAOA)**, the **Centre National des Opérations Aériennes (CNOA)**, the NATO **Combined Air Operations Centre (CAOC) / ARS Lyon-Mont Verdun**, and strategic nuclear airborne deterrent (Forces Aériennes Stratégiques - FAS) relay nodes (Ministère des Armées 2026 Order of Battle).

Location, Lithology & Structural Architecture: Located at 45°51'12"N, 4°47'06"E at the summit of Mont Verdun (elevation 626 m) in the Monts d'Or massif, Poleymieux-au-Mont-d'Or, 15 km northwest of Lyon, Rhône Department, France. Originally the site of an 1874 Séré de Rivières masonry fortress, the subterranean nuclear bunker ("Ouvrage enterré") was excavated between 1965 and 1970 inside Bajocian/Jurassic oolitic limestone beneath a 14-meter reinforced concrete blast slab and up to 250 meters of surrounding mountain limestone. The underground citadel comprises **16,000 m² (172,200 sq ft) across four subterranean stories** housed inside shock-isolated structural bays connected to the surface by hardened horizontal tunnels.

Equipment, Radar & C4ISR Systems: Topped by a 3D S-band/L-band Thales Ground Master 406 / ARES air defense radar dome on nearby Mont Thou; equipped with NATO ACCS LOC1 battle management consoles, Syracuse IV military satellite communication terminals, hardened diesel-electric power stations, chilled-water plants, and overpressure CBRN life support enabling 30 days of hermetically sealed operation for 600–800 command staff.

Ownership, Custodianship & Registered Officials: Owned by the **French Republic (Ministère des Armées)** and operated by the **Armée de l'Air et de l'Espace (French Air and Space Force)**. Key officials on record:

- **Général d'armée aérienne Jérôme Bellanger** — Chief of Staff of the French Air and Space Force (CEMAAE).
- **Général de corps aérien Vincent Cousin** — Commander of Air Defense and Air Operations (Commandant de la défense aérienne et des opérations aériennes - CDAOA).
- **Colonel Laurent / Base Commander, BA 942 Lyon-Mont Verdun** — Installation Commander, Base aérienne 942 "Capitaine Jean Robert".

#31. Heng Shan Military Command Center (衡山指揮所 / ROC Armed Forces JOCC)

Classification & Operational Purpose: Primary Wartime National Tri-Service Underground Command and Control Center of the Republic of China (Taiwan) Armed Forces (Heng Shan Command Center Record; Defense Engineering Analysis). Houses the **Republic of China Armed Forces Joint Operations Command Center (JOCC / 國軍聯合作戰指揮中心)**, serving as the wartime seat of the President of the Republic of China (Commander-in-Chief), the Minister of National Defense, and the Chief of the General Staff (Storm Media Technical Audit).

Location, Lithology & Structural Architecture: Located at 25°05'20"N, 121°32'40"E hollowed out inside **Jinan Mountain (雞南山)** in the Dazhi (大直) Military Fortress Zone, Zhongshan District, Taipei City, Taiwan. Initiated under President Chiang Kai-shek following the 1958 Second Taiwan Strait Crisis (823 Artillery Bombardment) and completed in 1982 after 15 years of tunneling. Excavated through Miocene sandstone and shale strata beneath 150–200 meters of rock overburden reinforced with multi-meter steel-plate and high-strength concrete liners designed to withstand conventional precision earth-penetrating munitions ("bunker busters"), 20-kiloton tactical nuclear overpressure, and electromagnetic pulse (EMP) strikes. Underground vehicular and pedestrian tunnels directly connect the complex to the Ministry of National Defense (MND) headquarters building, the Republic of China Air Force Command HQ, the Navy Command HQ, the **Yuan Shan Command Center (圓山指揮所)**, and the Songshan Airport / Heng Shan helipad evacuation nodes.

Equipment & C4ISR Systems: Houses the **Syun An (迅安) / Po Sheng (博勝) C4ISR data-link architecture** integrating Link-16 tactical feeds from E-2K Hawkeye AWACS, Pave Paws (Leshan Mountain) early-warning UHF phased-array radar, Tien Kung III / Patriot PAC-3 air defense batteries, and Hsiung Feng IIE/III missile commands; redundant underground power plants, NBC overpressure filtration, and joint U.S.-assessed structural hardening upgrades initiated in 2024–2025 (Storm Media 2024 Report).

Ownership, Custodianship & Registered Officials: Owned and operated by the **Ministry of National Defense (MND), Republic of China (Taiwan)** and the **General Staff Headquarters (參謀本部)**. Key officials on record:

- **Lai Ching-te** — President of the Republic of China and Commander-in-Chief of the ROC Armed Forces.
- **Wellington Koo Li-hsiung (顧立雄)** — Minister of National Defense, Republic of China.
- **Admiral Mei Chia-shu (梅家樹)** — Chief of the General Staff (參謀總長), ROC Armed Forces.

#32. CP TANGO (Command Post Theater Air Naval Ground Operations)

Classification & Operational Purpose: Primary Hardened Subterranean Wartime Theater Command Post for the **ROK-U.S. Combined Forces Command (CFC)**, **United States Forces Korea (USFK)**, and **United Nations Command (UNC)** (USFK Official Press Release March 2025; CP TANGO Technical Profile). Serves as the joint wartime C3I battle cab during annual *Ulchi Freedom Shield* and *Freedom Shield* theater exercises and in the event of hostilities on the Korean Peninsula (Korea JoongAng Daily).

Location, Lithology & Structural Architecture: Located at 37°25'38"N, 127°04'30"E carved into the northern granitic flank of Cheonggyesan mountain in Sucheong-dong, Sujeong-gu, Seongnam, Gyeonggi Province, South Korea (immediately south of Seoul and adjacent to Seoul Air Base [K-16]). Constructed in the 1970s inside solid Jurassic Daebo biotite granite under more than 200 meters of rock overburden, engineered to survive direct strikes from heavy bunker-busting artillery, ballistic missiles, and tactical nuclear weapons.

Equipment, SCIFs & Life Support Systems: Contains a multi-tiered amphitheater Combined Battle Cab with giant real-time common operational picture (COP) video walls, Sensitive Compartmented Information Facilities (SCIFs) directly linked via trans-Pacific submarine fiber and military satcom to the Pentagon, USINDOPACOM (Hawaii), and the National Reconnaissance Office (NRO), plus autonomous subterranean power generators, water purification plants, medical surgical suites, and closed-loop CBRN air regeneration supporting continuous sealed operation for up to two months without resupply.

Ownership, Custodianship & Registered Officials: Operated jointly by **United States Forces Korea (USFK)** and the **ROK-U.S. Combined Forces Command (CFC)** (USFK / CFC Official Roster). Key officials on record:

- **General Xavier T. Brunson, USA (succeeding Gen. Paul J. LaCamera)** — Commander, United Nations Command (UNC), ROK-U.S. Combined Forces Command (CFC), and United States Forces Korea (USFK).
- **General Kang Shin-chul, ROKA** — Deputy Commander, ROK-U.S. Combined Forces Command (CFC).

#33. Gyeryongdae Complex (계룡대 / Bunker B-2 & Tri-Service Underground HQ)

Classification & Operational Purpose: Integrated Tri-Service Headquarters of the Republic of Korea Armed Forces (ROK Army, ROK Navy, and ROK Air Force) and Hardened Subterranean Wartime Command Center (Gyeryongdae Complex Dossier). Designed to relocate South Korea's service headquarters southward from Seoul to reduce vulnerability to North Korean long-range artillery along the DMZ, while providing a deep mountain command bunker complementing the Ministry of National Defense's B-1 Bunker in Namtaeryeong, Seoul.

Location, Lithology & Structural Architecture: Located at 36°18'24"N, 127°14'15"E in Sindoan-myeon, Gyeryong City, South Chungcheong Province (Chungcheongnam-do), South Korea, nestled at the southern base of **Mount Gyeryongsan (845 m)**. Constructed between 1983 and 1989 under "Project 6-20" (89 事 業) across a 2,700-hectare restricted military reservation. Beneath the eight-story octagonal above-ground Tri-Service Headquarters building and inside the adjacent granitic bedrock of Mount Gyeryongsan lies a multi-level hardened underground command complex (designated within the ROK C2 network alongside the Capital Defense Command B-1 Bunker) carved into Mesozoic granite and reinforced with blast doors, EMP shielding, and underground vehicular tunnels.

Equipment & C4I Systems: Houses the **KJCCS (Korea Joint Command and Control System)**, Army Tactical Command Information System (ATCIS), Air Force MCRC (Master Control and Reporting Center) backup feeds, Navy KNCCS nodes, autonomous underground diesel power plants, and positive-pressure NBC life-support systems.

Ownership, Custodianship & Registered Officials: Owned and operated by the **Ministry of National Defense (MND) of the Republic of Korea** and the **Gyeryongdae Service Support Group (계룡대근무지원단)**. Key officials on record:

- **Kim Yong-hyun / Shin Won-sik** — Minister of National Defense, Republic of Korea.
- **Admiral Kim Myung-soo, ROKN** — Chairman of the Joint Chiefs of Staff (합동참謀議長), Republic of Korea Armed Forces.
- **General Park An-su (Chief of Staff, ROK Army); Admiral Yang Yong-mo (Chief of Naval Operations, ROK Navy); General Lee Young-su (Chief of Staff, ROK Air Force)** — Service Chiefs headquartered at Gyeryongdae.

#34. Mount Yamantau Facility (Mezhgorye / Beloretsk-15 & Beloretsk-16 / Ufa-105)

Classification & Operational Purpose: Deep Subterranean Strategic Continuity of Government (COG), Supreme Command Reserve Bunker, and Strategic Material Reserve Complex of the Russian Federation (Mount Yamantau Technical Dossier; TWZ Strategic Nuclear Command Analysis). Initiated in the late 1970s under Soviet Project "Metrostroy / US-30" and continuously expanded through the 1990s–2020s, the complex is supported by the closed administrative-territorial formation (ZATO) of **Mezhgorye** (unified in 1995 from the two closed military towns of *Beloretsk-15 / Tatly* and *Beloretsk-16 / Solnechny*, also designated *Ufa-105*).

Location, Lithology & Structural Architecture: Located at 54°15'18"N, 58°06'07"E inside **Mount Yamantau** (elevation 1,640 meters / 5,381 feet—the highest peak in the Southern Ural Mountains) within the Beloretsky District of the Republic of Bashkortostan, Russian Federation, surrounded by the restricted 252,000-hectare South Ural Nature Reserve. Excavated inside dense Proterozoic (Riphean) Zilmerdak and Mashak formation quartzitic sandstones and conglomerates under up to **800–850 meters (2,600–2,800 feet) of solid quartzitic rock overburden**. Because the high iron-ore and quartzite density of Yamantau attenuates direct through-mountain radio propagation, the complex serves primarily as an ultra-deep physical survival sanctuary for national leadership and strategic reserves (capable of sheltering up to 60,000 personnel across separated shock-isolated subterranean sectors linked by underground rail lines entering from the Beloretsk-15 [Southwest] and Beloretsk-16 [Northeast] portals).

Equipment & Engineering Infrastructure: Constructed by the **30th Directorate of Construction (FSUE US-30)**, a specialized deep-mining and metro-tunneling enterprise utilizing heavy tunnel boring machines, cast-iron/concrete tubbing liners, autonomous underground nuclear/diesel power stations, multi-stage regenerative air filtration plants, dedicated railway spurs connecting to the Inzer–Beloretsk main line, and redundant landlines linking to the Kosvinsky Kamen NC3 transmission hub.

Ownership, Custodianship & Registered Officials: Owned by the **Russian Federation (Ministry of Defense of the Russian Federation / Main Directorate of Special Programs of the President of the Russian Federation [GUSP])**, with civil administration handled by **ZATO Mezhgorye** and mining construction executed by **JSC "Upravleniye Stroitelstva No. 30" (US-30)**. Key officials on record:

- **Andrei R. Belousov** — Minister of Defense of the Russian Federation.
- **Alexander L. Linets** — Head of the Main Directorate of Special Programs of the President of the Russian Federation (GUSP — federal agency responsible for Russian presidential and government underground continuity facilities).
- **Viktor V. Panchenkov / Sergei N. Molchanov** — Head of the Administration of the Closed Administrative-Territorial Formation (ZATO) of Mezhgorye, Republic of Bashkortostan.

#35. Kosvinsky Kamen Complex (Kytlym / Object 1200 / Perimetr NC3 Node)

Classification & Operational Purpose: Hardened Subterranean Alternate Strategic Nuclear Command and Control (NC3) Center of the Russian General Staff and Central Command Post for the **15E601 "Perimetr" (Dead Hand)** Automated Retaliatory Nuclear Launch System (Kosvinsky Kamen Technical Profile; TWZ NC3 Analysis).

Location, Lithology & Structural Architecture: Located at 59°30'57"N, 59°03'43"E beneath **Mount Kosvinsky Kamen** (elevation 1,519 meters / 4,984 feet) near the settlement of Kytlym in the Karpinsk District of Sverdlovsk Oblast, Northern Ural Mountains, Russia. Excavated beginning in the late 1970s and completed in early 1996 inside a homogeneous Paleozoic Uralian Platinum Belt **dunite-pyroxenite-gabbro igneous pluton** beneath roughly **300 meters (1,000 feet) of solid crystalline granite/dunite overburden**. Crucially, the low-moisture, ultra-homogeneous crystalline rock of Kosvinsky Kamen permits **Very Low Frequency (VLF) and Extremely Low Frequency (ELF) radio waves to propagate directly through the mountain bedrock** from deeply buried subterranean antennas, preventing enemy nuclear strikes from blinding the facility by destroying surface antenna farms.

Equipment & Perimetr ("Dead Hand") NC3 Architecture: Houses the hardened computational core and command duty posts of the **Perimetr (15E601)** system, connected to an array of surface seismic geophones, radiation counters, atmospheric overpressure sensors, and early-warning radar feeds (Voronezh-DM). If activated during a strategic crisis and if sensors detect nuclear detonations on Russian soil combined with a loss of communications with the General Staff ("Kazbek" / "Cheget"), the Kosvinsky Kamen bunker issues automated launch commands to **15P011 command rockets (15A11 / 15Zh45 Sirena-M)** that broadcast launch-authorization codes across the territory of Russia directly to Strategic Rocket Forces (RVSN) ICBM silos, mobile launchers, and ballistic missile submarines.

Ownership, Custodianship & Registered Officials: Operated by the **Strategic Rocket Forces (RVSN)** and the **12th Chief Directorate of the Ministry of Defense of the Russian Federation (12 GUMO)** (supported by the adjacent Kytlym military garrison / Military Unit 20003). Key officials on record:

- **Colonel General Sergei V. Karakayev** — Commander of the Strategic Rocket Forces (RVSN) of the Russian Federation.
- **General of the Army Valery V. Gerasimov** — Chief of the General Staff of the Armed Forces of the Russian Federation.
- **Lieutenant General Igor A. Kolesnikov** — Chief of the 12th Chief Directorate (Nuclear Munitions & Special Facilities) of the Ministry of Defense of the Russian Federation.

#36. Objekt 1180 (Olișcani Soviet Reserve Command Bunker)

Classification & Operational Purpose: Unfinished Deep Subterranean Cylindrical Command Center of the Soviet Southwestern Strategic Direction High Command (Главное командование войск Юго-Западного направления) (Object 1180 Archival Record; Object 1180 Structural Survey). Construction began in 1985 at an estimated cost of 32 million Soviet rubles as one of a small class of standardized deep vertical-shaft Warsaw Pact theater command posts (sister facilities include Objekt 1161 in Kemelkowa, Belarus and the Bogdanovka bunker in Ukraine) before work was halted at ~85% structural completion upon the dissolution of the USSR in 1991.

Location, Lithology & Structural Architecture: Located at 47°48'29"N, 28°41'38"E inside a dense deciduous forest near the village of Olișcani, Șoldănești District, Republic of Moldova. Instead of horizontal mountain tunneling, Objekt 1180 consists of **two colossal vertical cylindrical reinforced-concrete shafts sunk 60 meters (197 feet) deep into the limestone/clay ground, each measuring 32 to 36 meters (105–118 feet) in diameter and divided into 12 subterranean stories** (with ceiling heights ranging from 4.5 m to 6 m). The two vertical cylinder blocks—Block A (Command & Computing Center) and Block B (Life Support, Diesel Power Plant, Chiller & Air Regeneration)—are capped at the surface by multi-meter reinforced concrete blast domes covered with earth mounds and linked underground by armored pedestrian and utility galleries.

Equipment & Present Condition: Retains heavy steel hermetic blast doors, vertical elevator and stairwell shafts, massive steel water/fuel storage tanks on the flooded lowest levels (Levels -10 to -12), and overhead bridge cranes, though electronic C4I gear was never installed prior to 1991 abandonment.

Ownership, Custodianship & Registered Officials: State property of the **Republic of Moldova** under the territorial jurisdiction of the **Ministry of Defense of the Republic of Moldova** and the Șoldănești District / Olișcani Commune Administration. Key officials on record:

- **Anatolie Nosatîi** — Minister of Defense of the Republic of Moldova.

#37. Bunker 703 (ChZ-703 / Special Archive of the USSR Ministry of Foreign Affairs)

Classification & Operational Purpose: Deep Subterranean Cold War Special Document Depository of the Ministry of Foreign Affairs of the USSR / Russian Federation, now operating as the **Museum of Modern Fortification (Bunker-703)** (PIR Center May 2026 Report; Bunker 703 Historical Dossier). Designed in the late 1940s under Lavrentiy Beria's special fortification program and commissioned in 1961 (Project ChZ-703) to store the Soviet Union's most sensitive original international treaties, diplomatic ciphers, and classified foreign policy archives until its declassification and vacating by the Foreign Ministry in 2005 due to deep groundwater seepage.

Location, Lithology & Structural Architecture: Located at 2nd Novokuznetsky Pereulok, Building 14, Structure 1, in the Zamoskvorechye district of central Moscow, Russia (55°43'55"N, 37°38'05"E), concealed behind the courtyard of an ordinary two-story 19th-century residential building adjacent to the Paveletskaya/Novokuznetskaya Metro lines. Excavated at a depth of **43 meters (141 feet) below street level** inside Carboniferous limestone and waterlogged clay using standard Soviet deep-metro tunneling techniques. The complex consists of a vertical elevator/stairwell shaft leading to two horizontal cylindrical tunnels (each **9.5 meters in diameter and ~40 meters long**) lined with bolted **cast-iron metro tubbing rings (чугунные тюбинги)** weighing 1.5 tons per segment, housing two-story internal steel archival vaults.

Equipment & Fortification Systems: Preserves working 10-ton hydraulic/electromechanical protective-hermetic blast doors (Защитно-герметические двери), Soviet FVU-series chemical/radiological air filtration columns, precision industrial dehumidification and climate-control plants designed to keep diplomatic paper archives at constant humidity, and a direct sealed service connection to the Zamoskvoretskaya Line of the Moscow Metro.

Ownership, Custodianship & Registered Officials: Owned by the **Russian Federation (Federal Agency for State Property Management [Rosimushchestvo])** and operated under lease since 2018 as a specialized fortification museum and archival research center by the **Autonomous Non-Profit Organization (ANO) "Center for the Study of Modern Fortification"**. Key officials on record:

- **Dr. Dmitry E. Yurkov (Дмитрий Евгеньевич Юрков)** — Historian, Author of *Soviet Secret Bunkers*, Founder and Scientific Director, Bunker-703 Museum (Associated Press / Bunker-703 Archive).

#38. Bunker 42 (Tagansky Protected Command Point / GO-42 / ChZ-293 / ChZ-572)

Classification & Operational Purpose: Deep Subterranean Cold War Reserve Command Post of the Soviet Long-Range Aviation (Дальняя авиация ВВС СССР) and Strategic Communications Hub, now operating as the **"Bunker-42 on Taganka" Cold War Museum and Exhibition Complex** (Tagansky Protected Command Point Record; Bunker-42 Official Portal). Initiated in 1947 under Stalin's personal decree (originally designated "Object-02", then **ChZ-293** in 1951 for the Ministry of Communications and **ChZ-572** in 1953 for Long-Range Aviation, and later **GO-42** in 1980), it entered combat duty in 1956—serving as the underground nuclear bomber command center during the 1962 Cuban Missile Crisis—before being decommissioned in 1995 and sold at a federal Rosimushchestvo auction in 2006 for 65 million rubles (Russpass Registry).

Location, Lithology & Structural Architecture: Located at 5th Kotelnichesky Pereulok, Building 11, Tagansky District, central Moscow, Russia (55°44'30"N, 37°38'54"E), disguised on the surface by a neo-classical yellow building with fake windows and a 6-meter concrete dome cap. Excavated by Metrostroy at a depth of **65 meters (213 feet) below ground** inside Carboniferous limestone directly alongside the Taganskaya (Koltsevaya Line) station of the Moscow Metro. The complex spans **7,000 m² (75,300 sq ft) across four parallel horizontal cylindrical blocks**, each **9.5 meters in diameter**, lined with heavy cast-iron metro tubbing and encased in steel-reinforced concrete grouting.

Equipment & Subterranean Systems: Block 1 and Block 2 housed the Long-Range Aviation Command Post, strategic bomber plotting boards, and telegraph/radio studio suites (from which Yuri Levitan had standby broadcast capability); Block 3 housed the central telegraph/communications switching exchange linking Moscow to Warsaw Pact capitals; and Block 4 housed twin submarine-grade diesel generators, artesian well water pumps (drawing from a 130-m borehole), and regenerative oxygen/CO_2 scrubber columns capable of sustaining 600–2,500 personnel for up to 30–90 days. Two sealed tunnels connect the bunker directly to the Taganskaya Metro station tunnels.

Ownership, Custodianship & Registered Officials: Privately owned and operated since 2006 by **OOO "Kompaniya Novik-Servis" (ООО «Компания Новик-Сервис»)**. Key officials on record:

- **Igor A. Kurylev (Игорь Алексеевич Курылев) / Alexei Kurylev** — Owner and General Director, OOO "Kompaniya Novik-Servis" / Bunker-42 on Taganka Museum Complex.

#39. Central Government War Headquarters (Burlington / Corsham / Site 3 / Turnstile)

Classification & Operational Purpose: Decommissioned Primary Cold War Continuity of Government (COG) Underground City of the United Kingdom (UK Ministry of Defence Official History of Corsham Tunnels; CGWHQ Dossier). Sequentially codenamed **Stockwell, Subterfuge, Burlington, Turnstile, Chanticleer, Peripheral, and Site 3**, this vast subterranean complex was secretively constructed between 1955 and 1962 inside former WWII aircraft engine manufacturing quarries (Spring Quarry) to serve as the wartime emergency seat of the British Prime Minister, the War Cabinet, the Chiefs of Staff, and 4,000 civil servants during a nuclear exchange (Subterranea Britannica Technical Monograph).

Location, Lithology & Structural Architecture: Located at 51°25'45"N, 2°12'18"W beneath the present-day **MoD Corsham** surface defence communications installation near Corsham, Wiltshire, England. Excavated at a depth of **30 to 37 meters (100 to 120 feet) underground** within the Jurassic Great Oolite (Bath Stone) limestone stratum. The underground city covers **97 hectares (240 acres / ~1 km²)**, divided into 22 self-contained structural areas linked by more than **100 kilometers (60 miles) of subterranean roadways and galleries** (named after London streets such as Piccadilly and Fleet Street), accessed via seven vertical personnel/freight shafts and two inclined haulage drifts.

Equipment & Subterranean Infrastructure:

- **Utilities & Command Suites:** Contained four huge subterranean power stations with 12 submarine-grade diesel generators cushioned on rubber shock mounts, an underground lake supplying treated water, the second-largest telephone exchange in Britain (GPO Exchange "Maiden" with 10,000+ internal lines and pneumatic Lamson message tubes linking every department), a BBC wartime radio broadcasting studio directly reserved for the Prime Minister, a subterranean hospital, bakery, laundry, kitchens, dormitories for 4,000 staff, and an underground railway spur connecting to the Box Tunnel on the Great Western Main Line so the Royal Train or Prime Minister's train could disembark directly inside the mountain. Declassified in December 2004 after being placed in care-and-maintenance in 1991.

Ownership, Custodianship & Registered Officials: Owned by the **United Kingdom Ministry of Defence (MoD)** and managed by the **Defence Infrastructure Organisation (DIO)** (while the surface complex above the quarry operates as the headquarters of **Defence Digital** at MoD Corsham). Key officials on record:

- **The Rt Hon John Healey MP** — Secretary of State for Defence, UK Ministry of Defence.
- **Mike Green** — Chief Executive, Defence Infrastructure Organisation (DIO), UK Ministry of Defence.

#40. Hack Green Bunker (Hack Green Secret Nuclear Bunker / RGHQ 10.2)

Classification & Operational Purpose: Decommissioned Royal Air Force ROTOR Radar Sector Operations Centre (M2 SOC) and Home Office Regional Government Headquarters (RGHQ 10.2 for North West England), now operating as the **Hack Green Secret Nuclear Bunker Museum** (Hack Green Official Portal; Cheshire Historical Dossier). Originally established in 1941 as a WWII Starfish decoy and Ground Controlled Interception (GCI) radar station, rebuilt in 1951–1955 under the secret **ROTOR program** as an underground concrete radar bunker, and completely reconstructed by the Home Office in 1981–1985 at a cost of £32 million as Regional Government Headquarters 10.2 before being declassified and sold in 1993.

Location, Lithology & Structural Architecture: Located at 53°02'48"N, 2°31'16"W off Baddington Lane near Nantwich, Cheshire, CW5 8AL, England. Consists of a semi-sunken and subterranean two-level heavily reinforced concrete blockhouse (measuring **3,300 m² / 35,000 sq ft**) with 3-meter-thick reinforced concrete outer walls, roof slab, and earth-bermed overburden.

Equipment & Museum Inventory: Retains the full RGHQ 10.2 Regional Commissioner's operations room, BBC emergency wartime broadcasting studio, BT Tornado communications exchange, original Luwa NBC air filtration plant, dual Ruston-Paxman diesel backup generators, decontamination showers, and one of the largest declassified collections of British nuclear weapons hardware in the UK (including deactivated casings and control consoles of **WE.177A/B/C** free-fall thermonuclear bombs, **Yellow Sun Mk.2**, **Blue Danube**, and a Cold War ballistic missile defense radar array) (Hack Green Technical FAQ).

Ownership, Custodianship & Registered Officials: Owned and operated since 1994/1998 by the **Hack Green Secret Nuclear Bunker Trust / Siebert Family**. Key officials on record:

- **Lucy Siebert** — Museum Director, Hack Green Secret Nuclear Bunker (Hack Green).
- **Rodney Siebert** — Founder, Principal Owner, and Chief Curator, Hack Green Secret Nuclear Bunker.

#41. Kelvedon Hatch Secret Nuclear Bunker (ROTOR R4 SOC / RGHQ 5.1)

Classification & Operational Purpose: Decommissioned Three-Level Underground RAF ROTOR Sector Operations Centre (R4 SOC) and Home Office Regional Government Headquarters (RGHQ 5.1 for Eastern England / London Transition), now operating as the **Kelvedon Hatch Secret Nuclear Bunker Museum** (Nuclear Essex / Kelvedon Hatch Official Portal; Subterranea Britannica Technical Survey). Built in 1952–1953 by Peter Lind & Co. after compulsory land purchase from the Parrish farming family to command the defense of the London Metropolitan Air Defence Sector; converted in the 1960s into a Civil Defence Regional Seat of Government (RSG 5), upgraded in the 1980s as RGHQ 5.1 for up to 600 personnel, decommissioned in 1992, and repurchased by the Parrish family in 1993.

Location, Lithology & Structural Architecture: Located at 51°40'12"N, 0°15'22"E off Kelvedon Hall Lane, Kelvedon Hatch, Borough of Brentwood, Essex, CM15 0LA, England. Constructed via "cut-and-cover" excavation inside a **38-meter (125-foot) deep pit dug into the London Clay** on a hillside. A three-story subterranean reinforced concrete box measuring **3,500 m² (38,000 sq ft)** with 3.05-meter (10-foot) thick tungsten-rod-reinforced concrete walls sits inside a shingle gravel blast-attenuation envelope topped by a 6-meter soil mound, accessed via a 110-meter (360-foot) dog-leg entrance tunnel disguised at the surface as a rural brick bungalow ("Kelvedon Hall Bungalow") (Kelvedon Hatch Structural Record).

Equipment & Subterranean Systems: Protected by 1.5-ton steel blast doors, positive-pressure NBC air conditioning plant (manufactured by Carrier Engineering Co.), two Rolls-Royce / Ruston diesel generators, BBC wartime radio studio, UK Warning and Monitoring Organisation (UKWMO) fallout plotting room, teleprinter banks, and government ministerial dormitories.

Ownership, Custodianship & Registered Officials: Privately owned and operated by **J.A. Parrish & Sons Ltd. / Nuclear Essex**. Key officials on record:

- **Mike Parrish (Michael Parrish)** — Owner and Managing Director, Kelvedon Hatch Secret Nuclear Bunker & J.A. Parrish & Sons Ltd. (LGfL Cold War Archive).

#42. Diefenbunker (CEGHQ Carp / Canadian Forces Station Carp)

Classification & Operational Purpose: Decommissioned Primary Federal Continuity of Government (COG) Complex of the Government of Canada, designated a **National Historic Site of Canada** in 1994 and operating since 1998 as **Diefenbunker: Canada's Cold War Museum** (Diefenbunker Official 25th Anniversary Report). Authorized in 1958 by Prime Minister John Diefenbaker under Project EASE (Experimental Army Signals Establishment) and operational from 1961 to 1994 as Canadian Forces Station (CFS) Carp, the bunker was designed to house 535 senior officials—including the Prime Minister, Governor General, Federal Cabinet, Chief of the Defence Staff, and Bank of Canada executives—for 30 days following a nuclear strike on Ottawa.

Location, Lithology & Structural Architecture: Located at 3929 Carp Road, Carp (West Carleton-March ward of Ottawa), Ontario, Canada (45°21'06"N, 76°02'50"W), 30 km west of Parliament Hill. Built via open-pit excavation inside a giant natural gravel hill: a **four-story, 35,000-metric-ton subterranean monolithic reinforced concrete structure spanning 9,300 m² (100,000 sq ft) across 358 rooms**. The structure is supported by 70 massive flared octagonal concrete columns (1.4 m diameter) engineered by foundation designer Dr. A.M. Lount to withstand a **5-megaton thermonuclear airburst at a distance of 1.8 kilometers (1.1 miles)**, cushioned on all sides and overhead by 1.5 meters of graded drainage gravel to absorb seismic ground shock waves.

Equipment & Subterranean Vaults:

- **Blast Tunnel & Bank of Canada Vault:** Accessed via a 115-meter (378-foot) open-ended horizontal blast dissipation tunnel (so a nuclear shockwave blows straight past the perpendicular entry doors without collapsing them), sealed by heavy hydraulic steel blast doors and decontamination showers. On the lowest level (Level 100) sits the **Bank of Canada Emergency Gold Vault**—a room-within-a-room secured by a 10-ton Mosler vault door designed to safeguard Canada's emergency gold reserves and paper currency plates during a nuclear war—alongside the CBC Emergency Broadcasting Studio, Federal Warning Centre, Prime Minister's Suite, surgical hospital, and three 400 kW Fairbanks-Morse submarine-style diesel generators.

Ownership, Custodianship & Registered Officials: Owned and operated by **Diefenbunker: Canada's Cold War Museum** (registered Canadian charitable non-profit museum corporation, transferred from the Municipality of West Carleton following DND decommissioning in 1994) (Diefenbunker Annual Report). Key officials on record:

- **Christine McGuire** — Executive Director, Diefenbunker: Canada's Cold War Museum (Official Biography).
- **Susan McLeod** — Chair of the Board of Directors, Diefenbunker: Canada's Cold War Museum.

#43. Bunker D-0 ARK (Atomska Ratna Komanda / Tito's Nuclear Bunker, Konjic)

Classification & Operational Purpose: Decommissioned Supreme Command Nuclear Bunker of the Socialist Federal Republic of Yugoslavia (SFRY) and Yugoslav People's Army (JNA), designated a National Monument of Bosnia and Herzegovina and operating as a Cold War Museum and Contemporary Art Biennial Venue (Tito's Bunker Official Portal; ARK D-0 Technical Record). Constructed in absolute secrecy over 26 years between March 1953 and September 1979 at a cost of **$4.6 billion USD** (one of the three most expensive military engineering projects in Yugoslav history, alongside Željava Underground Air Base [Objekt 505] and Lora Underground Naval Base) to shelter President Josip Broz Tito, his wife Jovanka, and the 350 highest political and military leaders of Yugoslavia for up to six months without external contact.

Location, Lithology & Structural Architecture: Located at 43°38'23"N, 17°59'41"E inside the southern flank of **Mount Zlatar** along the right bank of the Neretva River near Konjic, Herzegovina-Neretva Canton, Bosnia and Herzegovina (45 km southwest of Sarajevo). Disguised at the surface by three ordinary rural residential chalets connected via long corridors and triple 1.2-meter-thick steel nuclear blast doors to a horseshoe-shaped subterranean complex covering **6,854 m² (73,775 sq ft)** beneath a maximum vertical rock overburden of **280 meters (918 feet) of Triassic limestone and dolomite**, engineered to survive a direct **20-to-25-kiloton nuclear strike**.

Equipment & Twelve Functional Blocks: Divided into a "Protected Part" (horseshoe tunnel perimeter) and 12 internal structural blocks (Blocks 1–12): Block 8 (Marshal Tito's private luxury residence finished in Slavonian oak wall paneling), Block 6 (Supreme Command War Room and teleprinter/cipher center with 9 telephone exchanges), Block 9 (KLIMA Swiss/Yugoslav NBC air conditioning and filtration plant maintaining constant 21°C–23°C and 60%–70% RH), Block 10 (two 680 kVA Rade Končar / Jugoturbina diesel generators and a 50-ton subterranean oil tank), Block 11 (170 m³ fresh water treatment reservoir fed from the Neretva River), dormitory blocks, kitchens, and medical decontamination blocks. Saved from demolition in May 1992 when JNA officers Col. Šerif Mustafić and JNA personnel disconnected the detonation cables to 4.5 tons of pre-installed TNT charges during the Bosnian War.

Ownership, Custodianship & Registered Officials: Sovereign property of the **Ministry of Defence of Bosnia and Herzegovina (Ministarstvo odbrane BiH)**, with civilian management, preservation, and public tour operations entrusted to the municipal **Agency for Economic Development "Prvi korak" d.o.o. Konjic** (Bunker.ba Official Partner). Key officials on record:

- **Zukan Helez** — Minister of Defence of Bosnia and Herzegovina.
- **Osman Ćatić (Mayor of the City of Konjic) & Ajla Greda (Director, Agencija za ekonomski razvoj "Prvi korak" d.o.o. Konjic)** — Custodial municipal authorities managing Facility D-0 ARK.

#44. Dienststelle Marienthal (Regierungsbunker Ahrweiler / Ausweichsitz der Verfassungsorgane des Bundes)

Classification & Operational Purpose: Decommissioned Primary Federal Continuity of Government (COG) Bunker of the Federal Republic of Germany (West Germany), with its preserved eastern portal section operating since March 2008 as the **Dokumentationsstätte Regierungsbunker (Government Bunker Documentation Site)** (Dokumentationsstätte Regierungsbunker Official Portal; Ahrtal Official Record). Officially designated *Ausweichsitz der Verfassungsorgane des Bundes im Krisen- und Verteidigungsfall zur Wahrung von deren Funktionstüchtigkeit (AdVB)*, it was constructed between 1960 and 1972 at a cost of over DM 3 billion inside two never-completed WWI strategic railway tunnels (the *Kuxberg* and *Trotzenberg* tunnels along the unbuilt Strategischer Bahndamm) to house 3,000 federal constitutional officials—including the Federal President, Federal Chancellor, Cabinet, Joint Committee (Notstandsparlament), and Constitutional Court judges—for 30 days of sealed nuclear warfare.

Location, Lithology & Structural Architecture: Located at Am Silberberg 0, 53474 Bad Neuenahr-Ahrweiler, Rhineland-Palatinate, Germany (50°32'47"N, 7°04'38"E), 25 km south of the former West German capital of Bonn. Excavated inside the Silberberg and Kuxberg hills beneath up to **110 meters (360 feet) of Devonian Rhenish Massif graywacke slate (Grauwacke-Schiefer)**. The original underground complex comprised **17.3 kilometers (10.75 miles) of interconnected tunnels and four-story freestanding concrete-and-steel structures covering 83,000 m²**, divided into five autonomous sectors (Ost, West, Mitte, Nord, Süd) with 897 offices and 936 bedrooms.

Equipment, Deconstruction & Preserved 203-Meter Museum Section: Following a 1997 federal cabinet decision after the capital moved to Berlin, the complex was gutted between 2001 and 2006 for €16 million (removing the internal buildings to prevent groundwater contamination by asbestos and PCBs and sealing the deep rock tunnels behind concrete walls), leaving a fully intact **203-meter section of Bunker Ost (East Portal)** preserved as a federal monument. This preserved section retains the iconic **25-metric-ton hydraulic Mannesmann steel/concrete nuclear blast rolling gate (Rolltor)**, the decontamination sluice, Dräger NBC air filtration rooms, emergency medical operating rooms, and the Chancellor's / Presidential office reconstructions.

Ownership, Custodianship & Registered Officials: Owned by the **Federal Republic of Germany** via the **Bundesanstalt für Immobilienaufgaben (BImA)** (Federal Institute for Real Estate) and operated under long-term agreement by the **Heimatverein "Alt-Ahrweiler" e.V.** Key officials on record:

- **Executive Management Board (Vorstand), Bundesanstalt für Immobilienaufgaben (BImA)** — Federal property owner.
- **Karin Joachim** — Museum Director (Museumsleiterin), Dokumentationsstätte Regierungsbunker Bad Neuenahr-Ahrweiler (RegBu.de).

#45. Muskö Naval Base (Musköbasen / Musköვarvet)

Classification & Operational Purpose: Active Hardened Subterranean Naval Base, Underground Drydock Shipyard, and Fleet Command Headquarters of the Royal Swedish Navy (Svenska marinen) (The Guardian Defense Dossier; Muskö Naval Base Record). Constructed between 1950 and 1969 inside the granite cliffs of Muskö Island to protect Sweden's surface combatants and submarine fleet against nuclear strikes. After partial privatization in 2004, the Swedish Armed Forces officially relocated the **Swedish Naval Staff Headquarters (Marinstaben)** back into the underground complex in September 2019 alongside active naval repair operations by Saab Kockums AB.

Location, Lithology & Structural Architecture: Located at 58°59'28"N, 18°07'24"E on the island of Muskö in Haninge Municipality within the southern Stockholm Archipelago, Sweden (accessed from the mainland via the 3-km sub-sea Muskö Road Tunnel). Construction blasted **1,500,000 m³ (roughly**

4 million metric tons) of solid Precambrian granite bedrock beneath **40 to 65 meters (130 to 215 feet) of granite overburden**, creating an underground naval city encompassing **20 kilometers (12.4 miles) of subterranean roads**, barracks, a hospital, munitions magazines, workshops, and fleet command centers (Muskö Technical Profile).

Equipment & Three Subterranean Drydocks: Features **three giant rock-cut subterranean drydocks** accessible directly from the Baltic Sea through blast-protected sea portals cut into the granite cliff face: the largest drydock measures **350 meters (1,148 feet) long** (originally sized for *Halland*- and *Östergötland*-class destroyers and today servicing *Visby*-class stealth corvettes), while the two smaller docks (150 m and 80 m long) service *Gotland*-, *Södermanland*-, and *Blekinge*-class (A26) submarines. Each sea portal can be sealed against nuclear blast waves and overpressure by massive armored steel caisson/blast gates.

Ownership, Custodianship & Registered Officials: Owned by the **Kingdom of Sweden (Fortifikationsverket — Swedish Fortifications Agency)** and operated by the **Swedish Armed Forces (Försvarsmakten / Marinstaben)**, with the underground shipyard docks leased and operated by **Saab Kockums AB**. Key officials on record:

- **Rear Admiral (Konteramiral) Johan Norlén** — Chief of the Swedish Navy (Marinchef), Försvarsmakten (appointed 2024, succeeding Ewa Skoog Haslum).
- **Mats Wicksell** — Senior Vice President and Head of Business Area Kockums, Saab AB.
- **Maria Bredberg Pettersson** — Director General (Generaldirektör), Fortifikationsverket (Swedish Fortifications Agency).

#46. Haakonsvern Underground Naval Facility (Haakonsvern Orlogsstasjon)

Classification & Operational Purpose: Main Operational Base of the Royal Norwegian Navy (Sjøforsvaret), NATO North Atlantic Submarine Logistics Hub, and Hardened Subterranean Submarine Drydock and Munitions Complex (Haakonsvern Naval Base Record; NCC / Forsvarsbygg 2024 Submarine Dock Contract). Serves as the headquarters of the Royal Norwegian Navy, homeport for the Norwegian Fleet (Frigate and Submarine Squadrons), and a primary NATO maintenance hub undergoing a major NOK 4+ billion infrastructure expansion (2022–2029) to support the joint Norwegian-German **Type 212CD submarine** fleet.

Location, Lithology & Structural Architecture: Located at 60°20'16"N, 5°13'54"E at Mathopen on the Grimstadfjord, 15 km southwest of Bergen, Vestland County, Norway. Constructed between 1954 and 1962 (officially opened June 7, 1963) across a 107-hectare (1.07 km²) coastal and subterranean complex. Deep inside the surrounding Caledonian gneiss and granite mountains, massive rock caverns house an **underground drydock and maintenance hall connected directly to the fjord via a blast-protected sea tunnel**, alongside subterranean torpedo, naval mine, and Naval Strike Missile (NSM) magazines, fuel depots, underground workshops, a subterranean hospital, and hardened command and communications centers.

Equipment & Subterranean Shipyard Systems: The underground drydock cavern is equipped with heavy overhead bridge cranes, hull pressure-testing rigs, battery-charging and air-independent propulsion (AIP) maintenance bays, shock-isolated diesel power plants, and blast gates capable of sealing ships and submarines completely inside the mountain during wartime air or missile attacks.

Ownership, Custodianship & Registered Officials: Owned by the **Kingdom of Norway** via **Forsvarsbygg (Norwegian Defence Estates Agency)** and operated by the **Royal Norwegian Navy (Sjøforsvaret)**. Key officials on record:

- **Rear Admiral (Kontreadmiral) Oliver Berdal** — Chief of the Royal Norwegian Navy (Sjef Sjøforsvaret).
- **Flaggkommandør (Commodore) / Base Commander, Haakonsvern Orlogsstasjon** — Commander of Naval Base Haakonsvern.
- **Thorbjørn Thoresen** — Director General (Direktør), Forsvarsbygg (Norwegian Defence Estates Agency).

Section 8: Private Hardened Survival Complexes & Converted Strategic Missile Silos (#47–#50)

#47. Vivos Europa One (Rothenstein Complex, Thuringia, Germany)

Classification & Operational Purpose: Former East German National People's Army (NVA) Underground Munitions Fortress Re-Engineered as a Private Ultra-High-Net-Worth Luxury Survival Complex (Vivos Technical Record; Business Insider Engineering Dossier). Originally excavated by the Soviet military and the East German Ministry of National Defense (NVA Complex *Komplexlager 22 / Objekt Rothenstein*) inside the Trompeterfelsen mountain to store tactical missiles and strategic munitions; decommissioned by the German Bundeswehr and auctioned off in the 2000s before being acquired and marketed by The Vivos Group in 2015.

Location, Lithology & Structural Architecture: Located at 50°50'42"N, 11°36'08"E inside the Trompeterfelsen cliff overlooking the Saale River Valley at Rothenstein (near Jena), Saale-Holzland-Kreis, Thuringia, Germany. Excavated horizontally into Triassic Muschelkalk/Buntsandstein limestone and sandstone bedrock beneath up to **100 meters (328 feet) of rock overburden** across a 30.8-hectare (76-acre) fenced surface compound. The underground tunnel network encompasses **21,135 m² (227,904 sq ft) of subterranean chamber space** along more than **5 kilometers (3.1 miles) of vehicular-sized rock-cut tunnels** (plus an additional 4,078 m² / 43,906 sq ft of above-ground support and rail-transfer buildings).

Equipment & Hardening Specifications: Three main vehicular entry portals are sealed behind dual hydraulic nuclear blast and airtight doors, hardened against close-range nuclear airbursts, chemical/biological agents, F5-equivalent wind loads, and seismic shocks, equipped with redundant diesel power plants, deep-well water purification, and customizable two-story private residential suites (~230 m² / 2,500 sq ft each).

Ownership, Custodianship & Registered Officials: Operated and marketed by **The Vivos Group (Terravivos LLC)** (headquartered in Del Mar / Solana Beach, California, USA). Key officials on record:

- **Robert K. Vicino** — Founder, President, and Chief Executive Officer, The Vivos Group (Vivos Profile).

#48. Vivos xPoint (Former Black Hills Ordnance Depot, Edgemont, South Dakota)

Classification & Operational Purpose: World's Largest Private Hardened Bunker Community Comprising 575 Decommissioned U.S. Army Concrete Munitions Igloos (Vivos xPoint Record; 2026 Field Audit). Originally constructed in 1942 by the U.S. Army Corps of Engineers as the **Fort Igloo / Black Hills Ordnance Depot (BHOD)** to store munitions and bombs for up to 10,000 military and civilian personnel until base closure in 1967; leased and developed by The Vivos Group starting in 2016 as a private off-grid survival community for up to 5,000 residents.

Location, Lithology & Structural Architecture: Located at 43°13'22"N, 103°56'28"W at Igloo, 13 km southwest of Edgemont in Fall River County, southwestern South Dakota, USA. Spanning **46.6 km² (18 square miles / 11,520 acres)** of semi-arid prairie at an elevation of 1,150 m (3,800 ft), the complex contains **575 earth-bermed, steel-reinforced concrete barrel-arch bunkers ("igloos")** connected by 160 km (100 miles) of private roads. Each individual bunker measures either **8.0 m wide × 18.3 m long (26.5 ft × 60 ft = ~148 m² / 1,600 sq ft) or 8.0 m × 24.4 m (26.5 ft × 80 ft = ~204 m² / 2,200 sq ft)** with a 4.0-meter (13-foot) center arch height.

Equipment & Structural Hardening: Each arch bunker is constructed of high-strength reinforced concrete covered by 1.2 meters (4 feet) of compacted earth overburden and sealed by heavy steel blast doors, originally engineered by the U.S. Army to withstand an internal or external explosion generating **500,000 lbs (226,800 kg) of blast force (~35 psi overpressure) and 800 km/h (500 mph) shockwinds**. Individual leased bunkers (99-year leases) are outfitted by owners with off-grid solar/diesel microgrids, composting/septic tanks, NBC air filtration units, and water tanks fed by two deep artesian wells drawn from the Madison Limestone Aquifer.

Ownership, Custodianship & Registered Officials: Operated under master ground lease by **The Vivos Group (Vivos xPoint LLC)**. Key officials on record:

- **Robert K. Vicino** — Founder, President, and Chief Executive Officer, The Vivos Group (Vivos xPoint).

#49. Survival Condo (Converted Atlas-F ICBM Missile Silo, Cloud County, Kansas)

Classification & Operational Purpose: Decommissioned U.S. Air Force SM-65F Atlas Intercontinental Ballistic Missile Silo Converted into a 15-Story Subterranean Luxury Condominium Complex (Survival Condo Official Engineering Specification; Survival Condo Technical Dossier). Originally constructed in 1960–1962 by the U.S. Army Corps of Engineers at a cost of $15 million ($150M+ in 2026 USD) for the 550th Strategic Missile Squadron (Schilling Air Force Base) to house a 4-megaton nuclear-armed Atlas-F ICBM; decommissioned in 1965, acquired in 2008 by software engineer and defense contractor Larry Hall, and completely reconstructed by 2012 for $20 million into a self-contained vertical underground residential fortress.

Location, Lithology & Structural Architecture: Located at 39°25'15"N, 97°42'10"W north of Glasco / south of Concordia in Cloud County, north-central Kansas, USA. Consists of a **53.0-meter (174-foot) deep, 16.5-meter (54-foot) internal diameter vertical cylindrical shaft** encased in epoxy-hardened steel-reinforced concrete walls ranging from **0.76 meters (2.5 feet) to 2.74 meters (9 feet) thick**, capped by a monolithic concrete dome engineered to survive a nearby **12-megaton nuclear detonation (overpressure rating > 100 psi and 800 km/h winds)**. Inside the vertical silo, 15 structural steel floors provide **1,858 m² (20,000 sq ft) of floor space** (plus an adjacent subterranean Launch Control Center [LCC] spherical bunker).

Equipment, Redundant Life Support & Amenities:

- **Blast Doors & Five-Redundancy Utilities:** Sealed at the top entry by twin **7,250 kg (16,000-lb) armored steel blast doors**. Five-fold redundant power generation (local electrical grid, a surface 100 kW wind turbine, dual subterranean diesel generators with 38,000+ liters of fuel, and massive deep-cycle battery banks); triple-redundant 284,000-liter (75,000-gallon) subterranean water storage tanks with reverse-osmosis, UV, and carbon purification; military-grade nuclear, biological, and chemical (NBC) overpressure air filtration with blast valves; indoor hydroponic vegetable and aquaculture (tilapia) farms; an underground swimming pool, climbing wall, movie theater, medical bay, armory, and seven full-floor (170 m² / 1,820 sq ft) or half-floor (84 m² / 900 sq ft) luxury condominium residences equipped with LED virtual windows.

Ownership, Custodianship & Registered Officials: Owned and operated by **Survival Condo LLC** (and organized as a formal Kansas condominium association for unit owners) (Survival Condo Portal). Key officials on record:

- **Larry Hall** — Founder, Owner, Developer, and Chief Executive Officer, Survival Condo LLC (Survival Condo).

#50. The Oppidum (Kostelec nad Černými lesy / Vrchovany Complex, Czech Republic)

Classification & Operational Purpose: Decommissioned Late-Cold War Czechoslovak/Warsaw Pact Reserve Military Command Bunker Re-Engineered as the World's Largest Single-Owner Private Subterranean Ultra-HNWI Residence and Art Vault (Forbes Exclusive Technical Profile; Czech Architectural & Cadastral Dossier). Construction of the military bunker began in 1984 during heightened NATO-Warsaw Pact tensions and took 10 years to complete (finishing in 1994 after the Velvet Revolution); declared surplus by the Czech Army and acquired by Czech real estate entrepreneur Jakub Zamrazil, who partnered with former head of the Czech National Security Authority (NBÚ) Gen. Jiří Růžek and subterranean survival engineer Dr. Václav Cílek to transform the site into a private luxury fortress.

Location, Lithology & Structural Architecture: Located at 49°59'45"N, 14°52'10"E in a secluded forested valley near Kostelec nad Černými lesy / Český Brod in the Central Bohemian Region, Czech Republic (approximately 35 km east of Prague; a secondary northern site in Vrchovany was also evaluated during early planning) (Blesk Field Investigation). Encompasses a **30,000 m² (323,000 sq ft) walled surface estate** with a helipad and luxury administrative residence linked via a hidden tunnel and blast airlocks to a **two-level subterranean reinforced concrete bunker providing 7,200 m² (77,500 sq ft) of hardened underground living and vault space**.

Equipment & Autonomous Systems: Certified to NATO STANAG blast and CBRN protection standards, featuring hydraulic nuclear blast doors, closed-circuit CO_2 scrubbing and oxygen-regeneration systems (capable of up to 10 years of autonomous supply maintenance), subterranean diesel power generators, deep bedrock artesian water treatment, simulated circadian solar lighting, an underground spa and swimming pool, subterranean garden atrium, wine cellar, conference facilities, and CEN-certified high-security vaults for fine art and bullion.

Ownership, Custodianship & Registered Officials: Owned and developed by **Jakub Zamrazil / Oppidum Bunkers SA** (Swiss/Czech holding entity) (Czech Business Registry). Key officials on record:

- **Jakub Zamrazil** — Founder, Principal Owner, and Managing Director, The Oppidum / Oppidum Bunkers SA (Forbes).

Section 9: US Regional Subterranean Data Centers, Swiss Vaults, German Nuclear Mines & Taiwan Reserve C2 (#51–#59)

#51. InfoBunker (Former USAF/AT&T Cold War Bunker, Des Moines / Macksburg, Iowa)

Classification & Operational Purpose: Decommissioned Cold War Hardened Underground Communications Bunker Converted into a 100%-Uptime Subterranean Enterprise Colocation and Disaster Recovery Data Center (InfoBunker Official Technical Overview). Originally constructed during the Cold War as a nuclear-hardened subterranean coaxial/microwave communications switching center (designed to survive a 20-megaton nuclear blast within 8 km), acquired and rebuilt in 2005–2006 into an underground Tier III/IV-class commercial data center.

Location, Lithology & Structural Architecture: Corporate offices at 3101 Ingersoll Avenue, Des Moines, IA 50312; subterranean bunker complex located at 41°18'15"N, 94°08'40"W southwest of Des Moines near Macksburg / Winterset in Madison County, Iowa, USA (InfoBunker Location Profile). The entire **6,038 m² (65,000 sq ft) critical facility** sits completely underground inside a reinforced concrete shell featuring **0.91-meter (3-foot) thick steel-reinforced concrete ceilings and perimeter walls** buried beneath earth overburden, providing complete immunity to F5 tornadoes, windborne debris, and surface explosions.

Equipment, Power & Cooling Specifications:

- **2N+2 Redundant Power & HVAC:** Dual-path utility power backed by multiple subterranean diesel generators with 68,000+ liters (18,000 gallons) of underground diesel fuel storage, 2N+2 uninterruptible power supplies (UPS), and 2N+2 fully redundant subterranean air-handling and chilled-water cooling systems (InfoBunker Power & HVAC Specs). Secured by nuclear blast doors, biometric access mantraps, and diverse self-healing underground optical fiber rings.

Ownership, Custodianship & Registered Officials: Operated by **InfoBunker LLC / Five Nines Technology Group**. Key officials on record:

- **Executive Management Team** — Operations & Infrastructure Leadership, InfoBunker LLC / Five Nines (InfoBunker).

#52. Bluebird Underground Data Center (BBU, Springfield, Missouri)

Classification & Operational Purpose: Subterranean Limestone Cavern Colocation Data Center and Regional Optical Fiber Hub (Bluebird Fiber Official Data Center Specification; Bluebird Underground Engineering Dossier). Originally established by City Utilities of Springfield (SpringNet) and acquired by Bluebird Network in August 2014, followed by two major subterranean cavern expansions in 2016 and 2019–2021 (Data Center Dynamics).

Location, Lithology & Structural Architecture: Located at 2100 E. Trafficway Street, Springfield, Greene County, Missouri (37°13'12"N, 93°15'18"W). Positioned **26 meters (85 feet) vertically underground** inside the **Springfield Underground**—a 297,000 m² (3.2 million sq ft) decommissioned room-and-pillar limestone mine excavated into the Mississippian **Burlington-Keokuk Limestone** formation. Within the limestone cavern, Bluebird Underground occupies over **7,432 m² (80,000+ sq ft) of fully enclosed, climate-controlled concrete-and-steel data center buildings** (including a 29,500 sq ft Phase 2 expansion) supported by massive natural limestone pillars (Schneider Electric Technical Case Study).

Equipment, Power & Cooling Specifications: Dual 12 MW utility power feeds from separate City Utilities substations entering through dedicated vertical rock boreholes; N+1/2N Caterpillar diesel backup generators with underground fuel storage allowing 48+ hours of runtime at full load without refueling; Schneider Electric Symmetra MW UPS arrays; high-efficiency chilled-water CRAH units exploiting the constant 14°C–16°C (58°F–60°F) ambient limestone cavern temperature; SOC 2 Type II, HIPAA, and PCI-DSS certified security; and direct integration into Bluebird Fiber's 17,700+ km (11,000+ route-mile) Midwest fiber-optic backbone.

Ownership, Custodianship & Registered Officials: Owned and operated by **Bluebird Network, LLC (d/b/a Bluebird Fiber)** (backed by Macquarie Asset Management). Key officials on record:

- **Michael Morey / Johnie Johnson** — President and Chief Executive Officer, Bluebird Network / Bluebird Fiber (Bluebird Fiber).

#53. Cavern Technologies Data Center (LightEdge Lenexa Underground, Kansas)

Classification & Operational Purpose: Subterranean Limestone Cavern Enterprise Colocation and Private Data Suite Complex (LightEdge Lenexa Official Specification; Baxtel Acquisition Record). Founded in 2006 as Cavern Technologies to build customized, turn-key subterranean data centers inside the Meritex Lenexa Executive Park and acquired by LightEdge Solutions in September 2021 (DCD Expansion Report).

Location, Lithology & Structural Architecture: Located at 17501 W. 98th Street, Suites 18–39 (Pillar 25-33), Lenexa, Johnson County, Kansas 66219, USA (38°57'04"N, 94°47'28"W), 24 km southwest of downtown Kansas City. Situated **38.1 meters (125 feet) vertically underground** inside the **Meritex Lenexa Executive Park**—a 278,700 m² (3,000,000 sq ft) room-and-pillar subsurface complex excavated out of the Pennsylvanian Bethany Falls Limestone stratum. The data center encompasses over **18,580 m² (200,000+ sq ft) of hardened subterranean white and gray space** across two contiguous facilities (KC1 and KC2).

Equipment, Power & Environmental Specifications:

- **Geotechnical & Power Resilience:** Natural ambient rock temperature of 18°C–20°C (65°F–68°F) yields a low PUE and 30%–50% cooling energy reduction, while the 38-m limestone overburden eliminates F5 tornado, hail, lightning, and solar/EMP exposure. Equipped with 20+ MW of dual-

substation Evergy utility power, N+1/2N diesel generators, Liebert/Vertiv UPS and chilled-water loops, VESDA laser smoke detection with dual-interlock dry-pipe pre-action fire suppression, and ISO 27001, SOC 1/2/3, HIPAA, and PCI-DSS compliance.

Ownership, Custodianship & Registered Officials: Owned and operated by **LightEdge Solutions, Inc.** (backed by private equity firm GI Partners; subsurface industrial park owned by Meritex Enterprises, Inc.). Key officials on record:

- **Jim Masterson** — Chief Executive Officer, LightEdge Solutions, Inc. (LightEdge).
- **John Clune & James Kandt** — Co-Founders, Cavern Technologies (integrated into LightEdge Executive Leadership).

#54. Swiss Gold Safe Gotthard Vault (Gotthard Vault 1 & 2, Altdorf / Uri, Switzerland)

Classification & Operational Purpose: High-Security Private Alpine Bullion Depository, Bonded Warehouse (Offenes Zolllager - OZL), and Segregated Safe Deposit Vault Network (Swiss Gold Safe Official Technical Specification; Alpine Vault Engineering Profile). Founded in 2006 to provide bank-independent, off-balance-sheet segregated storage of physical gold, silver, platinum, palladium, diamonds, and cryptographic hardware keys inside the Gotthard Massif region.

Location, Lithology & Structural Architecture: Headquartered in Altdorf, Canton Uri, Switzerland (46°52'50"N, 8°38'22"E), operating five high-security vault facilities in the Gotthard Alpine region, including **Safe Deposit Box Vault Altdorf UR (Gotthard Vault 1 & Gotthard Vault 2)** and a fortified subterranean **Gotthard Alpine Bunker Bonded Warehouse** carved directly into the crystalline granite of the Saint-Gotthard Massif (Swiss Gold Safe Altdorf).

Equipment & Security Classifications: Constructed and certified to the highest European insurance security resistance classes (**EN 1143-1 VdS Resistance Grades X and XI**, including diamond-core-drill protection [CD]), equipped with seismic/acoustic wall sensors, biometric multi-factor mantraps, independent backup power, and 24/7 tactical monitoring connected directly to the Uri Cantonal Police (Kantonspolizei Uri). Operates a licensed Swiss Customs Bonded Warehouse (OZL) enabling white precious metals (silver, platinum, palladium) to be stored inside the Alpine bunker free of Swiss Value Added Tax (VAT).

Ownership, Custodianship & Registered Officials: Owned and operated by **Swiss Gold Safe AG** (Swiss Commercial Register CHE-113.025.816, headquartered at Bahnhofstrasse 28, 6460 Altdorf, Canton Uri) (Swiss Gold Safe Board). Key officials on record:

- **Ludwig Karl** — Member of the Board of Directors (Verwaltungsrat) and Spokesperson, Swiss Gold Safe AG (Official Executive Profile).
- **Board of Directors (Verwaltungsrat)** — Swiss Gold Safe AG, Altdorf, Canton Uri.

#55. Safe Host SH2 (STACK Infrastructure GVA02, Gland, Canton Vaud, Switzerland)

Classification & Operational Purpose: Hardened High-Security Enterprise and Hyperscale Data Center Complex (Invest Vaud Official Dossier; Data Center Dynamics Acquisition Report). Built between 2014 and May 2017 by Safe Host SA as the largest commercial data center in Western Switzerland (succeeding SH1 in Plan-les-Ouates, Geneva) and integrated in May 2022 into **STACK Infrastructure EMEA (as Facility GVA02)** following acquisition by IPI Partners (ICTjournal Technical Report).

Location, Lithology & Structural Architecture: Located at Avenue des Baumettes 23, 1196 Gland, District of Nyon, Canton Vaud, Switzerland (46°25'28"N, 6°16'04"E), along the Geneva–Lausanne A1 corridor. Constructed as a monolithic, windowless, multi-level subterranean and semi-subterranean reinforced-concrete bunker structure encompassing **14,000 m² (150,700 sq ft) of gross floor area** (with over 7,200 m² of high-density white technical space across four independent data center modules).

Equipment, Power & Security Specifications: Engineered to Uptime Institute Tier III and European EN 50600 Class 3/4 standards, delivering **16 MW to 20 MW of total critical IT power capacity** fed by redundant 125 kV/20 kV Romande Energie utility rings, 2N diesel rotary/static UPS plants, adiabatic free-cooling chillers achieving a PUE < 1.25, FINMA (Swiss Financial Market Supervisory Authority) circular compliance for sovereign and private banking workloads, and multi-perimeter anti-ram vehicle barriers and biometric security vaults (R&M Engineering Reference).

Ownership, Custodianship & Registered Officials: Owned and operated by **Safe Host SA / STACK Infrastructure EMEA** (acquired by IPI Partners in May 2022). Key officials on record:

- **Gérard Sikias** — Co-Founder and Former Chief Executive Officer, Safe Host SA (Techerati).
- **John Eland** — Chief Executive Officer, STACK Infrastructure EMEA.

#56. Olkiluoto Research Tunnel (VLJ Research Tunnel, Eurajoki, Finland)

Classification & Operational Purpose: Subterranean Hard-Rock Characterization and Full-Scale Engineered Barrier Testing Tunnel Adjacent to the Olkiluoto VLJ Repository (Posiva Oy Research Tunnel Record). Constructed in 1992–1993 prior to the excavation of the deeper ONKALO complex (#21) to conduct pioneering full-scale in-situ experiments on diamond wire-sawing, large-diameter vertical borehole drilling, bentonite buffer saturation, heater thermal-mechanical stress tests, and microbial/groundwater geochemistry in crystalline gneiss bedrock.

Location, Lithology & Structural Architecture: Located at 61°14'05"N, 21°26'25"E on Olkiluoto Island, Eurajoki, Finland, branching horizontally off the access ramp of the **VLJ Repository** (Voimalaitosjäte — TVO's operational bedrock repository for low- and intermediate-level reactor operational waste excavated at 60–100 meters depth). The Olkiluoto Research Tunnel is a **47-meter-long horizontal research drift excavated at a depth of 60 meters below surface** in Paleoproterozoic tonalitic gneiss and migmatite.

Equipment & In-Situ Experimental Systems: Contains three full-scale vertical experimental canister deposition holes (each **7.5 meters deep and 1.52 meters in diameter**, dry-bored using a rotary blind-hole boring machine with vacuum crushed-rock extraction), instrumented with electric canister heaters, piezoelectric acoustic emission transducers, fiber-optic temperature/strain gauges, and hydraulic bentonite swelling-pressure cells that validated the KBS-3V engineering specifications subsequently deployed at 430–455 m depth in ONKALO.

Ownership, Custodianship & Registered Officials: Owned by **Teollisuuden Voima Oyj (TVO)** and operated for scientific and repository engineering testing by **Posiva Oy**. Key officials on record:

- **Jarmo Tanhua** — President and Chief Executive Officer (Toimitusjohtaja), Teollisuuden Voima Oyj (TVO).
- **Ilkka Poikolainen** — President and Chief Executive Officer, Posiva Oy.

#57. Schacht Konrad (Endlager Konrad, Salzgitter, Lower Saxony, Germany)

Classification & Operational Purpose: Germany's First Licensed Under-Nuclear-Law Deep Geological Repository for Low- and Intermediate-Level Radioactive Waste with Negligible Heat Generation (BGE Endlager Konrad Official Portal; BGE Konrad Technical Monograph). Licensed under the German Atomic Energy Act (Planfeststellungsbeschluss of May 22, 2002, upheld by the Federal Administrative Court in March 2007) to permanently dispose of up to **303,000 m³ of radioactive waste with negligible heat generation** (representing roughly 95% by volume—though less than 1% by radioactivity—of all German nuclear waste from reactor decommissioning, medicine, industry, and research, plus waste retrieved from Asse II). Scheduled to complete underground conversion and enter active emplacement in the early 2030s.

Location, Lithology & Structural Architecture: Located at 52°11'02"N, 10°24'11"E in Bleckenstedt, Salzgitter, Lower Saxony, Germany. Originally sunk between 1957 and 1965 as Germany's deepest sedimentary iron-ore mine, extracting the 150-million-year-old Upper Jurassic (Oxfordian) **Coral Oolite (Korallenoolith)** iron ore horizon until mining ceased in 1976. The disposal horizon lies at an extraordinary depth of **800 to 1,300 meters (2,625 to 4,265 feet) below the surface**, sealed from upper groundwater aquifers by a **400-to-500-meter-thick impermeable Lower Cretaceous claystone barrier (Unterkreide-Tonstein)**, making Konrad one of the driest deep mines in Europe.

Equipment, Shafts & Emplacement Caverns:

- **Twin Vertical Shafts & 9 Emplacement Fields:** Served by two concrete/steel-lined vertical shafts spaced 1.8 km apart: **Shaft Konrad 1** (7.0 m diameter, 1,232 m deep—serving as personnel, material, and fresh-air intake shaft across Main Levels 2, 3, 4, 5, and 6 at 850 m to 1,200 m depth) and **Shaft Konrad 2** (7.0 m diameter, 999 m deep—rebuilt with a new 50-meter surface reloading hall and a 45-metric-ton heavy payload Koepe friction hoist exclusively for lowering standardized cubic/cylindrical nuclear waste containers and exhausting air).
- **Emplacement Chambers:** Up to 110,000 m³ of disposal chambers (7 m wide × 6 m high, up to 1,000 m long) are excavated in up to 9 emplacement fields using roadheaders, where shielded underground forklifts stack standardized Konrad steel/concrete waste containers before backfilling every void with shrinkage-compensated hydraulic pumping concrete.

Ownership, Custodianship & Registered Officials: Owned by the **Federal Republic of Germany** and constructed/operated by the 100% federally owned **Bundesgesellschaft für Endlagerung mbH (BGE)** (headquartered in Peine, supervised by the Federal Office for the Safety of Nuclear Waste Management [BASE] and the Federal Ministry for the Environment, Nature Conservation, Nuclear Safety and Consumer Protection [BMUV]) (BGE Management Board). Key officials on record:

- **Iris Graffunder** — Chairwoman of the Management Board (Vorsitzende der Geschäftsführung / CEO), Bundesgesellschaft für Endlagerung mbH (BGE) (BGE Official Imprint).
- **Marlis Koop & Dr. Thomas Lautsch / Jürgen Korth** — Managing Directors (Geschäftsführer), Bundesgesellschaft für Endlagerung mbH (BGE).
- **Christian Kühn** — President, Bundesamt für die Sicherheit der nuklearen Entsorgung (BASE).

#58. Asse II Mine (Schachtanlage Asse II, Remlingen, Lower Saxony, Germany)

Classification & Operational Purpose: Former Potash and Halite Salt Mine Utilized (1967–1978) as a Radioactive Waste Disposal Facility, Currently Undergoing Stabilization and Complete Nuclear Waste Retrieval under German Atomic Law (BGE Asse II Official Portal; World Nuclear News August 2025 Engineering Update). Between 1967 and 1978, the federally funded Gesellschaft für Strahlen- und Umweltforschung (GSF) emplaced **125,787 drums and containers of low-level radioactive waste (LLW) into 12 chambers on the 725-m and 750-m levels and 1,293 drums of intermediate-level radioactive waste (ILW) into Chamber 8a on the 511-m level** (totaling ~47,000 m³ of waste containing ~2.9 × 10^{15} Bq of activity, including ~28 kg of plutonium and ~102 tons of uranium). Following mounting rock-mechanical convergence and saturated NaCl/$MgCl_2$ brine inflows (~12.5 m³/day entering at the southern flank), the German Bundestag passed the 2013 **Lex Asse** amending § 57b of the Atomic Energy Act to mandate the complete retrieval (*Rückholung*) of all 126,000 drums from the mine.

Location, Lithology & Structural Architecture: Located at 52°07'46"N, 10°40'16"E near Remlingen in the Wolfenbüttel District of Lower Saxony, Germany (12 km southeast of Wolfenbüttel). Sunk between 1906 and 1908 into the steep **Asse Zechstein Salt Anticline (Asse-Salzsattel)**—a narrow diapiric ridge of Permian Zechstein rock salt (Leine and Staßfurt halite) and carnallite potash beds flanked by fractured Mesozoic overburden rocks. Between 1908 and 1964, commercial mining excavated **131 large chambers across 13 levels between 490 meters and 800 meters depth** (each chamber ~60 m long × 40 m wide × 15 m high), removing over 4 million m³ of salt and leaving thin pillar/flank barriers toward the water-bearing overburden (Asse II Geological History).

Equipment, Stabilization & Retrieval Shaft Asse 5:

- **Grouting, Brine Collection & Remote Retrieval:** To slow plastic convergence of the salt dome (~15 cm/year), BGE continuously pumps sorel cement (magnesium oxychloride concrete) into empty chambers while collecting and testing all inflowing overburden brine at the 658-m, 725-m, and 750-m levels. Because existing Shaft Asse 2 (765 m deep, 5.5 m diameter) lacks the hoisting capacity and nuclear containment rating required to extract retrieved waste packages, BGE and Redpath Deilmann are sinking a brand-new **800-meter-deep, 7.5-meter-diameter Retrieval Shaft (Schacht Asse 5)** 500 meters east of Shaft Asse 2, paired with a new underground retrieval mine horizon and remote-controlled/tele-operated robotic excavation machinery to open the 13 waste chambers starting in the 2030s (Redpath Deilmann Asse II Engineering Project).

Ownership, Custodianship & Registered Officials: Owned by the **Federal Republic of Germany** and operated since April 2017 by the **Bundesgesellschaft für Endlagerung mbH (BGE)** (which absorbed Asse-GmbH and BfS operational responsibilities) (BGE Asse II). Key officials on

record:

- **Iris Graffunder** — Chairwoman of the Management Board (Vorsitzende der Geschäftsführung), Bundesgesellschaft für Endlagerung mbH (BGE).
- **Dr. Thomas Lautsch** — Technical Managing Director overseeing Asse II Retrieval Engineering, BGE.

#59. Meishan Command Center (梅山營區 / Central-Southern Taiwan Underground Military C2)

Classification & Operational Purpose: Hardened Subterranean Regional and Reserve Military Command and Control (C2) Facility within the Republic of China (Taiwan) Armed Forces Defense Architecture (SCMP ROC Subterranean Backup C2 Dossier; Storm Media ROC Underground C2 Audit). Forms part of Taiwan's distributed subterranean theater command and backup C2 network (operating in coordination with the national **Heng Shan JOCC [#31]** in Taipei, the **Kunlun / Xinshe Command Center** of the ROC Army 10th Army Corps in central Taiwan, the **Lunshan / Qishan Command Center** of the 8th Army Corps in southern Taiwan, and the eastern underground air/naval bastions at **Jiashan [Hualien]** and **Chihhang [Taitung]**). Designed to preserve regional operational command, artillery/missile fire direction, and reserve mobilization across the Chiayi–Yunlin–Central Mountain Range axis if primary northern C2 nodes suffer kinetic degradation during a cross-strait conflict.

Location, Lithology & Structural Architecture: Located in the Meishan / Dalin foothill sector of Chiayi County, Taiwan (23°35'00"N, 120°33'00"E) along the western approaches to the Alishan / Central Mountain Range fault-thrust belt. Constructed into Miocene-Pliocene sandstone and mudstone foothills with reinforced concrete portal tunnels, blast-baffle dog-leg corridors, and subterranean command rooms isolated from surface fragmentation and artillery/cruise-missile overpressure.

Equipment & C4ISR Integration: Linked into the ROC Armed Forces fiber-optic and microwave tactical backbone (IMSE / Syun An C4ISR network), equipped with independent diesel generator sets, subterranean fuel and water cisterns, NBC overpressure air filtration, and hardened VHF/UHF/HF radio and mobile satellite relay terminals supporting the **ROC Army 10th Army Corps (陸軍第十軍團指揮部)** and Southern Theater Reserve Brigades.

Ownership, Custodianship & Registered Officials: Owned and operated by the **Ministry of National Defense (MND), Republic of China (Taiwan)** and the **Republic of China Army (ROCA 10th Army Corps / Fifth Theater of Operations)**. Key officials on record:

- **Wellington Koo Li-hsiung (顧立雄)** — Minister of National Defense, Republic of China (Taiwan).
- **General Chung Shu-ming (鍾樹明)** — Commander of the Republic of China Army (陸軍司令).
- **Lieutenant General Chang Yuan-hsun (章元勳) / Commander, ROC Army 10th Army Corps** — Operational Theater Commander overseeing the Central-Chiayi Defense Sector (Fifth Theater of Operations).

Section 10: Categorized Primary Archival, Statutory, Engineering & Scientific Sources

In strict compliance with forensic and technical verification standards, all data points, structural metrics, geological formations, and institutional leadership registries in this 59-facility audit are grounded in the following primary statutory, governmental, regulatory, and scientific record classes:

I. Defense, Continuity of Government & Statutory Archives

- **United States Department of Defense (DoD) & USNORTHCOM / NORAD:** NORAD & USNORTHCOM Official Fact Sheet: Cheyenne Mountain Complex; USNORTHCOM Cheyenne Mountain Portal; U.S. Space Force Space Base Delta 1 Leadership Roster; U.S. Government Accountability Office (GAO) Report GAO-09-994R: Full Costs and Security Implications of Cheyenne Mountain Realignment.
- **Washington Headquarters Services (WHS) & Pentagon Force Protection Agency (PFPA):** Department of Defense Instruction (DoDI) 5110.11: Raven Rock Mountain Complex (RRMC); WHS FY 2022 Military Construction Budget Justification (Pentagon Reservation / Raven Rock Mountain Complex); Pentagon Force Protection Agency Annual Report.
- **U.S. Code of Federal Regulations (FEMA Mount Weather & OPM Boyers):** 44 CFR Part 15 — Conduct at the Mt. Weather Emergency Assistance Center and at the National Emergency Training Center; 44 CFR § 15.2 Statutory Definitions (Mt. Weather Executive Director); U.S. Office of Personnel Management (OPM): 200 Feet Underground at Boyers, PA.
- **Allied & European Defense Ministries:** UK Ministry of Defence: Corsham Tunnels (Burlington Bunker) Official History; French Ministry of the Armed Forces: Base aérienne 942 Lyon-Mont Verdun / CDAOA; U.S. Forces Korea (USFK): Command Post Theater Air Naval Ground Operations (CP TANGO); Norwegian Defence Estates Agency (Forsvarsbygg) / NCC Haakonsvern Submarine Infrastructure; Diefenbunker: Canada's Cold War Museum Official Impact & Governance Report; Dokumentationsstätte Regierungsbunker Bad Neuenahr-Ahrweiler (BImA); PIR Center Dossier on Bunker 703 (Special Archive of the USSR MFA).

II. Sovereign Monetary Depositories, Central Banks & Ecclesiastical Archives

- **United States Mint & Federal Reserve Bank of New York:** United States Mint: Fort Knox Bullion Depository Specification; U.S. Mint Historical & Structural Dossier on Fort Knox; Federal Reserve Bank of New York: 33 Liberty Street Gold Vault Engineering & Custody Specification; FRBNY President John C. Williams: The Gold Beneath Our Feet.
- **European Central Banks, BIS & Vatican Apostolic Archive:** London Bullion Market Association (LBMA): Bank of England Threadneedle Street Gold Vaults; Bank for International Settlements (BIS) Basel Management & Governance; Swiss National Bank (SNB) Governing Board & Bern Bundesplatz 1 Head Office; Archivum Apostolicum Vaticanum: Historical & Technical Notes on the Cortile della Pigna Bunker; Official Appointment of Prefect Rev. Prof. Rocco Ronzani, O.S.A.

III. Nuclear Regulatory Authorities & Deep Geological Repository Operators

- **Posiva Oy (Finland):** Posiva Oy: Research and Final Disposal Facilities at ONKALO; ONKALO Principle of Operation & KBS-3V Engineering; Posiva Oy Executive Leadership Appointment.
- **U.S. Department of Energy (DOE) & Nuclear Regulatory Commission (NRC):** U.S. DOE Waste Isolation Pilot Plant (WIPP) Portal; DOE Carlsbad Field Office (CBFO) & SIMCO Leadership; U.S. NRC Review of DOE's Yucca Mountain Site Characterization & Exploratory Studies Facility (ESF); National Academy of Engineering: Safety in the Underground Construction and Operation of the Exploratory Studies Facility at Yucca Mountain.
- **Bundesgesellschaft für Endlagerung mbH (BGE, Germany):** BGE Endlager Schacht Konrad Portal; BGE Technical Monograph: Endlager Konrad; BGE Schachtanlage Asse II Retrieval Portal; BGE Management Board Registry.

IV. Deep Underground Physics Laboratories, Biobanks & Commercial Subsurface Operators

- **Astroparticle Physics Institutes:** INFN Laboratori Nazionali del Gran Sasso (LNGS) Directorate; ICRR University of Tokyo: Super-Kamiokande & Kamioka Observatory; SNOLAB Annual Report & Executive Directorate; Tsinghua University & Yalong River Hydropower: China Jinping Underground Laboratory (CJPL-I & CJPL-II); INR RAS: Baksan Neutrino Observatory (BNO) Technical Specification; Sanford Underground Research Facility (SURF) & SDSTA Executive Leadership.
- **Biobanks, Archival Vaults, Data Centers & Alpine Engineering:** Crop Trust & NordGen: Svalbard Global Seed Vault; Piql AS & Arctic World Archive Foundation STI; Granite Mountain Records Vault (FamilySearch International); Hunt Midwest SubTropolis Specification; Underground Vaults & Storage (UV&S) Hutchinson Salt Mine; Implenia & SBB Gotthard Base Tunnel Lot 360 (MFS Sedrun); Bahnhof Pionen White Mountain Data Center; LightEdge / Cavern Technologies Lenexa; Bluebird Underground Data Center; InfoBunker Des Moines; Swiss Gold Safe AG Gotthard Vaults; Survival Condo Atlas-F ICBM Engineering.